

# FDG PDF / Org / Typst Audit

This standalone file is a starter audit comparing three artifacts:

- Published PDF: `fdg-book/fdg_book.pdf`
- Upstream Org/source material: `fdg-book/scheme/org/*.org`
- Current Typst book output: `fdg-book.pdf`

The goal is not to claim full fidelity yet. The goal is to make every partial comparison visible, so later audit runs can turn each partial or unchecked row into an exhaustive row without losing the evidence trail.

Project decisions recorded after the MVP:

- Letter layout is intentional; page-for-page reproduction of the published PDF is not a goal.
- Errata in the current book PDF is intentional project end matter.
- `typ/main.pdf` and `fdg-book.pdf` are expected to differ only by cover/index material.
- This audit should remain a standalone Typst document.
- Reference numbers should match the published PDF where possible. Current implementation seeds citation order invisibly from the published reference list, while keeping generated bibliography output.

## Audit Contents

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## Audit Labels

|                  |                                                                                           |
|------------------|-------------------------------------------------------------------------------------------|
| <b>KNOWN</b>     | Evidence is concrete enough to rely on for future work.                                   |
| <b>PARTIAL</b>   | Some representative source/PDF/Typst evidence has been checked, but not the whole region. |
| <b>UNCHECKED</b> | Inventory exists, but no meaningful comparison has been completed yet.                    |
| <b>RISK</b>      | Likely fidelity gap or known difference that needs action.                                |
| <b>TARGET</b>    | What a later exhaustive pass should produce.                                              |

Audit entry statuses are color-coded separately from coverage labels:

|                           |                                                                                    |
|---------------------------|------------------------------------------------------------------------------------|
| <b>EXACT</b>              | Checked exhaustively for the stated region and matches the chosen target.          |
| <b>PARTIAL-EXACT</b>      | Sampled or scoped evidence matches, but the claim is not exhaustive.               |
| <b>ACCEPTED-STYLISTIC</b> | Intentional style/layout difference; not a bug unless content is obscured or lost. |
| <b>SOURCE-DRIFT</b>       | Org/source differs from the published PDF or chosen canonical text.                |

|                           |                                                                              |
|---------------------------|------------------------------------------------------------------------------|
| <b>PDF-DERIVED-REPAIR</b> | Typst intentionally repairs or reconstructs content from the published PDF.  |
| <b>PROJECT-ADDED</b>      | Intentional project-specific material outside the published PDF body target. |
| <b>BUG</b>                | Actionable defect in conversion, source, layout, or output.                  |
| <b>UNCHECKED</b>          | Not meaningfully checked yet.                                                |

## Inputs and Build Facts

| Artifact         | Path                      | Observed metadata                                                   | Audit note                                                                                                                                             |
|------------------|---------------------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Published PDF    | fdg-book/fdg_book.pdf     | 249 pages; 432 x 639.36 pt; Acrobat Distiller; modified 2017-01-16. | Canonical visual/text target for book body and published index.                                                                                        |
| Current book PDF | fdg-book.pdf              | 170 pages; US letter; Typst 0.15.0.                                 | Generated by <code>make book</code> ; includes intentional cover/index/errata material.                                                                |
| Draft PDF        | typ/main.pdf              | 167 pages; US letter; Typst 0.15.0.                                 | Generated by <code>make draft</code> ; expected to differ from <code>fdg-book.pdf</code> only by cover/index material.                                 |
| Org source       | fdg-book/scheme/org/*.org | 19 Org files at top level.                                          | <code>fdg.org</code> currently includes only preface, prologue, and chapter 1; the converter/book assembly uses the individual content files directly. |
| Typst source     | typ/content/*.typ         | 18 content files.                                                   | Generated from Org plus repo-specific repairs and presentation support.                                                                                |

Reproduction commands used for this MVP:

```
pdftinfo fdg-book/fdg_book.pdf
pdftinfo fdg-book.pdf
pdftinfo typ/main.pdf
pdftotext -f 1 -l 8 fdg-book/fdg_book.pdf -
pdftotext -f 1 -l 8 fdg-book.pdf -
pdftotext -f 230 -l 249 fdg-book/fdg_book.pdf -
pdftotext -f 160 -l 170 fdg-book.pdf -
rg -n '^(\s+|#fdg-chapter)' typ/content/*.typ
rg -n '^\\s*+' fdg-book/scheme/org/*.org
git log --date=short --pretty=format:'%h %ad %s' -n 20
```

## Executive Diff

| Area            | Published PDF                                               | Org source                                         | Current Typst PDF                                                | MVP finding                                                                                                                                        |
|-----------------|-------------------------------------------------------------|----------------------------------------------------|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Physical layout | 249 pages, small custom page size.                          | No final pagination.                               | 170 pages, letter paper.                                         | <b>KNOWN</b> Letter layout is intentional. Use semantic/content checks first; layout differences are accepted unless they obscure or drop content. |
| Front matter    | LaTeX/MIT Press title, copyright, Einstein quote, contents. | <code>fdg.org</code> has title metadata and quote. | Custom Typst cover, title, copyright, quote, generated contents. | <b>PARTIAL</b> Main matter is aligned in order; typogra-                                                                                           |

|                |                                                                      |                                                                                     |                                                                                    |                                                                                                                                                                       |
|----------------|----------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                |                                                                      |                                                                                     |                                                                                    | phy and contents pagination differ.                                                                                                                                   |
| Body chapters  | Preface, prologue, chapters 1-11, appendices A-C, references, index. | Matching chapter files exist, except <code>fdg.org</code> is not a complete driver. | Matching Typst content files exist and are included by <code>typ/main.typ</code> . | <b>PARTIAL</b> Existing notes say every chapter/appendix has at least one published-PDF spotcheck; preface, prologue, chapters 1-2, and chapter 11 had deeper passes. |
| End matter     | References then index.                                               | <code>references.org</code> and <code>errata.org</code> exist.                      | References, generated index, then errata.                                          | <b>KNOWN</b> Errata is intentionally included. The generated index is intentional, but still needs content/locator audit.                                             |
| References     | Published extraction begins with SICP as [1].                        | Bib file follows published order.                                                   | Current extraction begins with SICP as [1] after citation-order seeding.           | <b>KNOWN</b> Numbering/order now matches sampled published references. Formatting details still need exhaustive audit.                                                |
| Figures        | Figures embedded in published PDF.                                   | Figure mentions in Org.                                                             | 8 extracted PDF assets used in Typst.                                              | <b>PARTIAL</b> Figure inventory exists; visual placement/caption fidelity is not yet exhaustive.                                                                      |
| Code blocks    | Typeset Scheme listings and results.                                 | Org Scheme/result blocks.                                                           | Typst generated fenced Scheme blocks and result text.                              | <b>PARTIAL</b> Known converter repairs fixed misclassified prose and cached result placeholders in early chapters. Need systematic code/result comparison.            |
| Math/equations | Published equations are canonical when Org is empty or drifted.      | Some Org equation environments are empty or drift from PDF.                         | Typst contains repaired/reconstructed equations.                                   | <b>PARTIAL</b> Chapter 11 equations 11.1-11.14 and 11.20-11.21 were reconstructed from PDF according to existing notes.                                               |

## Structural Coverage Ledger

| Region  | Org -> Typst   | PDF -> Typst   | Current status                                                                              |
|---------|----------------|----------------|---------------------------------------------------------------------------------------------|
| Preface | <b>PARTIAL</b> | <b>PARTIAL</b> | Deep pass mentioned in <code>PDF-FIDELITY-NOTES.md</code> ; needs line-level audit entries. |

|                |                |                |                                                                                                                                                   |
|----------------|----------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Prologue       | <b>PARTIAL</b> | <b>PARTIAL</b> | Deep pass mentioned; Scheme/prose classification fixes already landed.                                                                            |
| Chapter 1      | <b>PARTIAL</b> | <b>PARTIAL</b> | Deep pass mentioned; Cartan noweb placeholder and cached-result placeholder issues fixed.                                                         |
| Chapter 2      | <b>PARTIAL</b> | <b>PARTIAL</b> | Deep pass mentioned; coordinate-independence wording/formula drift fixed.                                                                         |
| Chapters 3-10  | <b>PARTIAL</b> | <b>PARTIAL</b> | At least one PDF page spotchecked per chapter; needs exhaustive section-by-section comparison.                                                    |
| Chapter 11     | <b>PARTIAL</b> | <b>PARTIAL</b> | Deep pass mentioned; many equations reconstructed from PDF.                                                                                       |
| Appendices A-C | <b>PARTIAL</b> | <b>PARTIAL</b> | At least one PDF page spotchecked per appendix; Appendix C end sample shows semantic match but reference-number phrasing differs.                 |
| References     | <b>PARTIAL</b> | <b>PARTIAL</b> | Published order is encoded in <b>fdg-reference-order</b> ; sampled numbering now matches. Formatting and every entry still need exhaustive audit. |
| Index          | <b>PARTIAL</b> | <b>PARTIAL</b> | Both PDFs contain indexes, but generated entries/page numbers differ substantially.                                                               |
| Errata         | <b>KNOWN</b>   | <b>KNOWN</b>   | Intentional project end matter. Compare against Org, not against the published PDF body target.                                                   |

## Known Findings to Preserve

These come from `PDF-FIDELITY-NOTES.md` and should be retained as historical audit evidence, even if future passes re-check the same regions.

- **KNOWN** 2026-07-02 spotcheck: one published-PDF page from each generated chapter and appendix was compared against Typst.
- **KNOWN** Chapter 4 equation 4.3: missing  $x_k$  subscript was restored from the published PDF.
- **KNOWN** Chapter 7 equation 7.9: punctuation was moved out of `\mathsf{g}` so the final factor renders as **g**, followed by punctuation.
- **KNOWN** Preface/prologue/chapter 1/chapter 2: a deeper pass fixed misclassified wrapped parenthetical prose, expanded **Cartan**, removed non-published cached-result placeholders, and repaired chapter 2 wording/formula drift.
- **KNOWN** Chapter 11: a deeper pass reconstructed equations 11.1-11.14 and 11.20-11.21 from the PDF because the Org source has empty equation environments there.
- **KNOWN** Chapter 11: fixes also covered **Einstein**, Lorentz interval terms, aligned Lorentz displays, `\bar.v.double` norm notation, `make-4-tuple`, and  $R^{\scriptscriptstyle (= 1)}$ .
- **KNOWN** `scripts/assert-typst-regressions.mjs` preserves automated regression coverage for generated Typst structure, selected PDF text expectations, figures, labels, and bibliography keys. As of this audit pass it reports: 15 targeted files, 1 PDF text file, 18 generated files, 74 global checks, 8 figures, 599 labels, and 21 bibliography keys.

## MVP Findings

### A-001: Published PDF and Typst PDF use different layout targets **KNOWN**

The published PDF is 249 pages at 432 x 639.36 pt. The current book PDF is 170 pages at US letter. This means page-number comparisons are useful only inside each artifact; they are not a direct fidelity metric between artifacts.

**TARGET** Later exhaustive layout work should audit for dropped, obscured, or badly placed content inside the letter layout. It should not attempt to reproduce the published page geometry.

### A-002: Book assembly is not the same as **fdg.org** **KNOWN**

`fdg-book/scheme/org/fdg.org` includes only:

```
#+include: preface.org :minlevel 1
#+include: prologue.org :minlevel 1
#+include: chapter001.org :minlevel 1
```

The current Typst assembly in `typ/main.typ` includes preface, prologue, chapters 1-11, appendices A-C, references, generated index, and errata. Therefore, the meaningful Org comparison target is the set of individual content files, not `fdg.org` alone.

**TARGET** Future converter/audit tooling should record exactly which Org file produced each Typst file, including generated or manually repaired regions.

#### A-003: Current PDF intentionally includes extra project end matter **KNOWN**

The published PDF end matter is references followed by index. The current Typst PDF table of contents shows references, index, and “Errata for FDG.” This is an intentional project addition, not a published-PDF fidelity bug.

**TARGET** Future audit rows should compare errata against `fdg-book/scheme/org/errata.org` and `typ/content/errata.typ`, while marking it out of scope for published-PDF body matching.

#### A-004: Reference order is seeded to match the published PDF **KNOWN**

Sampled published PDF references begin:

```
[1] Harold Abelson and Gerald Jay Sussman with Julie Sussman, Structure and Interpretation of Computer Programs
[2] Harold Abelson and Andrea deSessa, Turtle Geometry
```

Before the fix, sampled current Typst PDF references began:

```
[1] H. Abelson and A. deSessa, Turtle Geometry
[2] S. A. Papert, Mindstorms
...
[14] H. Abelson, G. J. Sussman, and J. Sussman, Structure and Interpretation of Computer Programs
```

That was a content-level difference, not just layout, because it changed citation numbers throughout the book.

The current Typst references are still generated with:

```
#bibliography("../references.bib", title: none, full: true, style: "ieee")
```

The numbering is now forced into published order by `fdg-reference-order` and `fdg-seed-bibliography-order()` in `typ/fdg-lib/refs.typ`, called from the cover/title page so it does not add pages. Verification samples from `fdg-book.pdf` after the change:

```
Turtle Geometry [2]
IEEE Standard for the Scheme Programming Language [10]
Spivak, Calculus on Manifolds [17]
References: [1] Structure and Interpretation of Computer Programs
```

**TARGET** Exhaustive pass should compare every citation callout and bibliography entry against the published PDF. Remaining expected differences are bibliography formatting/style details, not numbering/order.

#### A-005: Index exists in both PDFs but is not equivalent **PARTIAL**

The published PDF has a long index beginning after references, including symbol entries such as composition, derivative notation, tuple notation, and Scheme syntax entries. The current Typst PDF has a generated index with different entry set, sort behavior, page numbers, and visible generated-entry wording.

**TARGET** A full index audit should classify entries into: present in both, published-only, Typst-only, same term/different locator, and locator unavailable because page geometry intentionally differs.

#### A-006: Figures are inventoried but not visually audited PARTIAL

Current Typst uses these extracted assets:

```
fig-2-1.pdf
fig-2-2.pdf
fig-2-3.pdf
fig-4-1.pdf
fig-4-2.pdf
fig-5-1.pdf
fig-6-1.pdf
fig-7-1.pdf
```

The Org and Typst text mention the same figure numbers in the sampled regions. No exhaustive visual check has yet verified scaling, placement, caption punctuation, or page adjacency against the published PDF.

**TARGET** Later runs should screenshot or rasterize each figure in both PDFs, compare caption text, and mark whether visual differences are acceptable Typst styling or content loss. The imported PDF assets are serviceable for the current build, but the likely long-term direction is to redraw the figures natively in CeTZ so figure geometry, labels, text layers, and captions are under Typst source control rather than inherited from embedded PDF crops.

#### A-007: Chapter 11 depends on PDF-derived repairs KNOWN

Chapter 11 is a special case because the Org source contains empty equation environments for a run of published equations. The Typst source now includes reconstructed equations 11.1-11.14 and 11.20-11.21, so a naive Org-vs-Typst diff will report intentional additions.

**TARGET** Mark PDF-derived repairs inline in generated Typst or in a machine-readable sidecar, so future converter runs do not erase them silently.

### Chunk 1 Audit Entries: Front Matter and Global Rules

| ID     | Region            | Status             | Finding                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------|-------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C1-001 | Artifact metadata | ACCEPTED-STYLISTIC | Published PDF is 249 pages at 432 x 639.36 pt. Current book PDF is 170 US-letter pages. Draft PDF is 167 US-letter pages. Letter layout is intentional; page-for-page reproduction is out of scope.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| C1-002 | Cover/title       | ACCEPTED-STYLISTIC | Published PDF begins with a simple title leaf and title page. <code>fdg-book.pdf</code> begins with a custom Typst cover, then a title page with title/authors/MIT Press location text. <code>typ/main.pdf</code> uses a draft title leaf instead of the custom cover. This is intentional.                                                                                                                                                                                                                                                                                                                                                                                   |
| C1-003 | Copyright/CIP     | PARTIAL-EX-ACT     | Current book and draft PDFs preserve the copyright, Creative Commons license text, MIT Press sales paragraph, Computer Modern/LaTeX statement, Library of Congress data, ISBN, subject lines, classification, control number, and print line. Source of current rendering: <code>typ/fdg-lib/title.typ:40-69</code> ; caller: <code>typ/main.typ</code> title setup. Published evidence: front matter pages from <code>pdftotext -f 1 -l 10 fdg-book/fdg_book.pdf -</code> ; current evidence: <code>pdftotext -f 1 -l 10 fdg-book.pdf -</code> . Extraction differs in punctuation/ligatures/underscore handling, but no content loss was found in the sampled front matter. |
| C1-004 | Einstein quote    | ACCEPTED-STYLISTIC | The quote text and attribution are present. Published PDF renders visible quotation marks around the passage; current Typst uses a quote block without literal enclosing quotation marks. Treat as accepted styling unless visual inspection later shows ambiguity.                                                                                                                                                                                                                                                                                                                                                                                                           |

|        |                |                    |                                                                                                                                                                                                                                                                                           |
|--------|----------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C1-005 | Contents       | ACCEPTED-STYLISTIC | Published contents is less granular and uses published pagination. Current Typst contents is generated from headings, includes more subsections/exercises, and uses Typst page numbers. Ordering is coherent and no missing top-level region was found in the sampled contents.           |
| C1-006 | Page numbering | ACCEPTED-STYLISTIC | Current Typst sets front matter to roman numerals after title pages, then resets chapter 1 to Arabic page 1. Absolute and roman page numbers differ from the published PDF because layout and contents granularity differ intentionally.                                                  |
| C1-007 | Draft vs book  | KNOWN              | typ/main.pdf has 167 pages and starts with a draft title leaf. fdg-book.pdf has 170 pages and starts with the custom cover. The expected difference is cover/index material; both include the intentional errata material.                                                                |
| C1-008 | Errata policy  | PROJECT-ADDED      | Errata appears in the current Typst outputs and is intentional project end matter. It should be compared against fdg-book/scheme/org/errata.org and typ/content/errata.typ, not treated as published-PDF body drift.                                                                      |
| C1-009 | Index policy   | PROJECT-ADDED      | fdg-book.pdf includes a generated index; typ/main.pdf omits it. Published PDF includes an index, but page locators cannot match exactly because page geometry differs intentionally. Future index audit should compare entry coverage and locator policy rather than exact page equality. |

Chunk 1 front-matter follow-up findings:

| ID     | Region                | Status | Finding / requested correction                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------|-----------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C1-010 | License banner        | BUG    | Current rendering is built in typ/fdg-lib/title.typ:40-69; the license paragraph is at :43. The copyright page is missing the Creative Commons CC BY-NC-SA banner/mark from the desired front matter treatment. Repair entry point: add the mark/banner in title.typ near the license paragraph, then visually compare current fdg-book.pdf front matter against published PDF front matter pages. No implementation change made in this audit pass. |
| C1-011 | MIT Press sales email | BUG    | Current source is typ/fdg-lib/title.typ:47. special_sales@mitpress.mit.edu should be set as raw/monospace text; creativecommons.org in title.typ:43 should remain in the main text font, which is currently correct. Repair should be local to the sales paragraph, likely using Typst raw/monospace markup for only the email address.                                                                                                              |
| C1-012 | LaTeX logo            | EXACT  | typ/fdg-lib/title.typ:49 uses #LaTeX; typ/fdg-lib/title.typ:1 imports it from the metalogo package via basics.typ. The rendered stylistic baseline offsets are correct and should be preserved.                                                                                                                                                                                                                                                      |
| C1-013 | CIP heading spacing   | BUG    | Current source is typ/fdg-lib/title.typ:51-52: only #v(1em) precedes Library of Congress Cataloging-in-Publication Data. There should be additional vertical space before this heading. Repair entry point: adjust the vertical spacing in title.typ, then visually compare the copyright/CIP page.                                                                                                                                                  |
| C1-014 | CIP p. cm. line       | BUG    | Current source is typ/fdg-lib/title.typ:56. The p. cm. line should be indented relative to surrounding Cataloging-                                                                                                                                                                                                                                                                                                                                   |

|        |                            |     |                                                                                                                                                                                                                                                                                                       |
|--------|----------------------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        |                            |     | in-Publication lines. Repair entry point: wrap or isolate that line in <code>title.typ</code> with the desired indentation rather than changing generated content.                                                                                                                                    |
| C1-015 | Print line                 | BUG | Current source is <code>typ/fdg-lib/title.typ:67-68</code> . The final print-number line <code>10 9 8 7 6 5 4 3 2 1</code> should have wider spacing between numbers than the current Typst rendering. Repair entry point: use explicit spacing or a small table/sequence in <code>title.typ</code> . |
| C1-016 | Classification punctuation | BUG | Current source is <code>typ/fdg-lib/title.typ:63</code> , which has <code>516.3'6--dc23</code> . It should use an em dash before <code>dc23</code> , matching the published/CIP typography more closely. Repair is a local punctuation change in <code>title.typ</code> .                             |
| C1-017 | Control number alignment   | BUG | Current source is <code>typ/fdg-lib/title.typ:65</code> . <code>2012042107</code> should be right-aligned on the copyright/CIP page. Repair entry point: align just this line in <code>title.typ</code> and visually compare the front matter page.                                                   |

Chunk 1 evidence commands:

```
pdftotext -f 1 -l 10 fdg-book/fdg_book.pdf -
pdftotext -f 1 -l 10 fdg-book.pdf -
pdftotext -f 1 -l 8 typ/main.pdf -
pdfinfo fdg-book/fdg_book.pdf
pdfinfo fdg-book.pdf
pdfinfo typ/main.pdf
sed -n '1,140p' typ/fdg-lib/title.typ
sed -n '1,120p' typ/main.typ
```

## Chunk 2 Audit Entries: References and Citations

| ID     | Region             | Status         | Finding                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------|--------------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C2-001 | Reference order    | EXACT          | Published PDF, <code>fdg-book/scheme/org/references.org:4-64</code> , <code>typ/references.bib:1-156</code> , and <code>typ/fdg-lib/refs.typ:12-34</code> now agree on the 21-entry order: SICP is [1], Turtle Geometry [2], Bishop/Goldberg [3], Carroll [4], Church [5], Flanders [6], Frankel [7], Galilei [8], Hawking/Ellis [9], IEEE Scheme [10], MTW [11], Pais [12], Papert [13], Schutz [14], Singer/Thorpe [15], Spivak Comprehensive [16], Spivak Calculus [17], Sussman/Wisdom Role [18], SICM [19], Wald [20], software [21]. Citation seeding implementation lives at <code>typ/fdg-lib/refs.typ:36-43</code> . |
| C2-002 | Citation numbering | PARTIAL-EX-ACT | Sampled current PDF callouts match the published numbering after citation-order seeding: Turtle Geometry [2], Galileo Galilei [8], SICM [19], Spivak Calculus [17], Bishop/Goldberg [3], MTW/Carroll/Schutz/Wald [11]/[4]/[14]/[20], IEEE Scheme [10]. A later pass should mechanically check every citation callout.                                                                                                                                                                                                                                                                                                         |
| C2-003 | Bibliography style | BUG            | Current visible bibliography source is <code>typ/content/references.typ:5-6</code> , generated by <code>scripts/convert-org-to-typst.mjs:1215-1217</code> , using <code>#bibliography("../references.bib", title: none, full: true, style: "ieee")</code> . This preserves numbering but renders IEEE-style prose, not the published bibliography prose from <code>fdg-book/scheme/org/references.org:4-64</code> . Repair entry point: either change bibliography style machinery or implement the man-                                                                                                                      |

|        |                         |           |                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------|-------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        |                         |           | ual visible bibliography prototype in C2-F02/C2-F05 while keeping hidden bibliography/citation resolution.                                                                                                                                                                                                                                                                                                                           |
| C2-004 | Author names            | BUG       | Entry data locations: [1] is fdg-book/scheme/org/references.org:4-5 and typ/references.bib:1-7; [19] is Org :58-59 and BibTeX :138-144. Current bibliography abbreviates most authors to initials and renders and where published prose uses with Julie Sussman and with Meinhard E. Mayer. Repair likely requires manual published-prose rendering, because BibTeX author lists cannot express these with phrases using IEEE style. |
| C2-005 | Entry [6]               | BUG       | Published [6] is fdg-book/scheme/org/references.org:19-20. BibTeX entry is typ/references.bib:39-46, with note = {Dover, New York, 1989} at :45. Current IEEE output drops the note. Repair entry point: custom/manual visible bibliography or style support that renders note for this entry.                                                                                                                                       |
| C2-006 | Entry [10]              | BUG       | Published [10] is fdg-book/scheme/org/references.org:30-31. BibTeX entry is typ/references.bib:68-73, with organization at :70 and note/standard number at :72. Current IEEE output renders only the title/year and loses the standard number/organization. Repair entry point: manual visible bibliography or adjusted entry/style mapping for manuals.                                                                             |
| C2-007 | Entry [18]              | BUG       | Published [18] is fdg-book/scheme/org/references.org:54-56. BibTeX entry is typ/references.bib:129-136. Current output quotes the title, says technical report AIM-2002-18, and abbreviates the month; published prose says Artificial Intelligence Laboratory memo AIM-2002-018, November 2002. Repair should preserve published wording exactly in a manual visible bibliography or custom renderer.                               |
| C2-008 | Entry [21]              | BUG       | Published [21] is fdg-book/scheme/org/references.org:63-64. BibTeX entry is typ/references.bib:153-156. Current output renders an IEEE online citation; published prose reads Free software is available at: followed by the URL. Repair likely requires manual visible bibliography text.                                                                                                                                           |
| C2-009 | Publisher/address order | BUG       | For many book entries, published prose uses Publisher, Address, Year; current IEEE output often uses Address: Publisher, Year or separates sentences. Examples: [1], [2], [3], [11], [15], [16], [17], [19].                                                                                                                                                                                                                         |
| C2-010 | Title styling           | UNCHECKED | Published bibliography visually italicizes book/report titles in the source/PDF. pdftotext cannot verify styling; a visual PDF inspection or screenshot comparison should decide whether generated bibliography title styling is acceptable.                                                                                                                                                                                         |

Chunk 2 reference-entry map:

| No. | BibTeX key        | Published lead                      | Current status               |
|-----|-------------------|-------------------------------------|------------------------------|
| 1   | abelson1996sicp   | SICP, full names with Julie Sussman | BUG author style             |
| 2   | abelson1980turtle | Turtle Geometry                     | BUG author/punctuation style |
| 3   | bishop1968tensor  | Tensor Analysis on Manifolds        | BUG publisher/address style  |

|    |                         |                                                                 |                                  |
|----|-------------------------|-----------------------------------------------------------------|----------------------------------|
| 4  | carroll2003spacetime    | Spacetime and Geometry                                          | <b>BUG</b> author style          |
| 5  | church1941calculi       | The Calculi of Lambda-Con-<br>version                           | <b>BUG</b> author style          |
| 6  | flanders1963forms       | Differential Forms... + Dover<br>note                           | <b>BUG</b> missing rendered note |
| 7  | frankel1997geometry     | The Geometry of Physics                                         | <b>BUG</b> author style          |
| 8  | galilei1623assayer      | Il Saggiatore                                                   | <b>BUG</b> author style          |
| 9  | hawking1973large        | The Large Scale Structure of<br>Space-Time                      | <b>BUG</b> hyphen/title style    |
| 10 | ieee1991scheme          | IEEE Std 1178-1990                                              | <b>BUG</b> missing standard/org  |
| 11 | misner1973gravitation   | Gravitation                                                     | <b>BUG</b> author/address style  |
| 12 | pais1982subtle          | Subtle is the Lord                                              | <b>BUG</b> author style          |
| 13 | papert1980mindstorms    | Mindstorms                                                      | <b>BUG</b> author style          |
| 14 | schutz1985first         | B. Schutz A First Course...                                     | <b>BUG</b> punctuation/style     |
| 15 | singer1967topology      | Lecture Notes on Elemen-<br>tary Topology and Geome-<br>try     | <b>BUG</b> author/address style  |
| 16 | spivak1970comprehensive | A Comprehensive Introduc-<br>tion to Differential Geome-<br>try | <b>BUG</b> author/address style  |
| 17 | spivak1965calculus      | Calculus on Manifolds                                           | <b>BUG</b> author/address style  |
| 18 | sussman2002role         | Role of Programming memo                                        | <b>BUG</b> report wording        |
| 19 | sussman2001sicm         | SICM, with Meinhard E.<br>Mayer                                 | <b>BUG</b> author style          |
| 20 | wald1984general         | General Relativity                                              | <b>BUG</b> author style          |
| 21 | fdg-software            | Free software is available at                                   | <b>BUG</b> URL prose             |

Chunk 2 potential fix notes, not yet implemented:

| ID     | Idea                               | Status                         | Initial finding                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------|------------------------------------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C2-F01 | Custom CSL/bibli-<br>ography style | <b>UNCHECKED</b>               | A custom CSL-style route may improve author-name and punctuation choices while preserving generated bibliogra-<br>phy machinery. Unknown risk: exact published quirks such<br>as with Julie Sussman, Free software is available at:,<br>the [6] Dover note, and the [18] memo wording may be<br>difficult or impossible to express cleanly in CSL. Needs a<br>small prototype before choosing.                                                                                                                                                                                                                                                                             |
| C2-F02 | Manual visible bibli-<br>ography   | <b>PARTIAL-EX-<br/>ACT</b>     | Most exact route is to render the visible references<br>from a structured published-order list or directly from<br>fdg-book/scheme/org/references.org:4-64, while keeping<br>Typst citations resolved through a hidden gener-<br>ated bibliography. Current generated visible source is<br>only typ/content/references.typ:5-6; generator branch<br>is scripts/convert-org-to-typst.mjs:1215-1217. A local<br>test showed that @abelson1996sicp fails without any bibli-<br>ography, but compiles as [1] when #bibliography("typ/<br>references.bib", ...) is hidden in a zero-sized box. The<br>hidden bibliography did not appear in extracted PDF text<br>in that test. |
| C2-F03 | Generator-owned<br>solution        | <b>ACCEPTED-<br/>STYLISTIC</b> | If manual visible bibliography is chosen, it should<br>be generated by scripts/convert-org-to-typst.mjs or a                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|        |                             |           |                                                                                                                                                                                                                                                                                                                                                        |
|--------|-----------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        |                             |           | helper data structure, not hand-edited in <code>typ/content/references.typ</code> , because that file is regenerated from Org.                                                                                                                                                                                                                         |
| C2-F04 | Citation order seed         | EXACT     | <code>fdg-reference-order</code> and <code>fdg-seed-bibliography-order()</code> already solve numbering/order. Any future bibliography fix should preserve this behavior or replace it with an equally explicit published-order mechanism.                                                                                                             |
| C2-F05 | Recommended first prototype | UNCHECKED | Prototype a manual visible <code>fdg-published-bibliography()</code> that renders the 21 entries exactly enough to match <code>references.org</code> , plus a hidden <code>#bibliography</code> solely for citation resolution. Verify page count, visible reference text, citation callouts, and that no hidden references leak into text extraction. |

Chunk 2 evidence commands:

```
pdftotext fdg-book/fdg_book.pdf /private/tmp/fdg-published-all.txt
pdftotext fdg-book.pdf /private/tmp/fdg-current-all.txt
sed -n '11320,11375p' /private/tmp/fdg-published-all.txt
sed -n '11165,11230p' /private/tmp/fdg-current-all.txt
rg -n '@[A-Za-z][A-Za-z0-9_-]+' typ/content/*.typ
sed -n '1,220p' typ/references.bib
sed -n '1,180p' fdg-book/scheme/org/references.org
sed -n '1,90p' typ/fdg-lib/refs.typ
typst compile - /private/tmp/no-bib-test.pdf
typst compile - /private/tmp/hidden-bib-test.pdf
pdftotext /private/tmp/hidden-bib-test.pdf -
```

### Chunk 3 Audit Entries: Preface, Prologue, Chapters 1-2

| ID     | Region                             | Status         | Finding                                                                                                                                                                                                                                                                                                                                                                     |
|--------|------------------------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C3-001 | Chunk scope                        | PARTIAL-EX-ACT | PDF-FIDELITY-NOTES.md records a deeper pass through the preface, prologue, chapter 1, and chapter 2 on 2026-07-02. This chunk preserves those prior repairs and adds fresh spot findings from the published PDF text, Org source, generated Typst source, and current <code>fdg-book.pdf</code> . It is not yet a line-by-line exhaustive pass.                             |
| C3-002 | Preface structure                  | PARTIAL-EX-ACT | The preface body, <code>Acknowledgements</code> heading, author signature, location, and date are present in current Typst output. The sampled acknowledgement citations now resolve to the published numbers: Turtle Geometry [2] and Mindstorms [13].                                                                                                                     |
| C3-003 | Preface coverage                   | PARTIAL-EX-ACT | No new preface content loss was found in this chunk beyond citation/reference items already tracked elsewhere. Detailed quote typography and similar visual style checks are intentionally out of scope for this pass.                                                                                                                                                      |
| C3-004 | Prologue prose/code classification | PARTIAL-EX-ACT | Prior repairs prevented wrapped explanatory prose from being misclassified as Scheme blocks. Sampled current output around the Lagrange-equation discussion now reads as prose and math rather than accidental code; representative Scheme listings such as <code>Lagrange-equations</code> , <code>Gamma</code> , and <code>oscillator</code> examples remain code blocks. |
| C3-005 | Prologue footnote citation         | BUG            | Published footnote 1 ends can be found in Papert [13]. Org source location: <code>fdg-book/scheme/org/prologue.org:288-291</code> includes the literal terminal [13]. Generated location: <code>typ/content/prologue.typ:7</code>                                                                                                                                           |

|        |                              |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------|------------------------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        |                              |                | drops it, and current <code>fdg-book.pdf</code> ends the footnote at plain <code>Papert</code> . This is a converter/reference-handling bug inside footnotes, not an upstream Org omission. Likely repair entry points: citation replacement in <code>scripts/convert-org-to-typst.mjs:1160-1165</code> and footnote/main split handling around <code>scripts/convert-org-to-typst.mjs:152-156</code> . Fix should translate this source reference to <code>@papert1980mindstorms</code> or otherwise preserve the generated citation in the footnote.                                                                                                                                                       |
| C3-006 | Prologue source drift        | SOURCE-DRIFT   | Both Org and current Typst say for mass <code>m</code> and spring constant <code>k</code> . this lagrangian is implemented by with lowercase <code>this</code> and lowercase <code>lagrangian</code> . Because the same wording appears in the upstream Org, this is not a converter regression, but it should be compared against the published PDF visually/textually before deciding whether to preserve source or repair to published prose.                                                                                                                                                                                                                                                             |
| C3-007 | Chapter 1 opening/citation   | PARTIAL-EX-ACT | The chapter 1 Galileo epigraph and citation resolve to [8], matching the published PDF and Org source. No additional typography check is requested for this row.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| C3-008 | Chapter 1 degree glyphs      | BUG            | Published PDF extraction renders both longitude mentions as <code>90° E</code> , with spacing before <code>E</code> . Org source location: <code>fdg-book/scheme/org/chapter001.org:43-44</code> has mixed glyphs ( <code>90°E</code> , then <code>90°E</code> ). Generated location: <code>typ/content/chapter001.typ:14</code> preserves the mixture and omits the space before <code>E</code> . Repair entry point: source normalization in <code>scripts/convert-org-to-typst.mjs</code> targeted repairs or upstream Org cleanup, followed by visual check of the chapter 1 opening page.                                                                                                               |
| C3-009 | Chapter 1 Cartan repair      | PARTIAL-EX-ACT | Prior notes say the noweb Cartan placeholder was expanded and standalone cached-result placeholders not in the published PDF were removed. Org source around <code>fdg-book/scheme/org/chapter001.org:455-464</code> still uses the noweb <code>&lt;&lt;Cartan&gt;&gt;</code> pattern and cached result marker. Generated output at <code>typ/content/chapter001.typ:208-219</code> contains the explanatory paragraph and explicit <code>(define Cartan (Christoffel-&gt;Cartan ...))</code> block. Converter repair entry point: <code>scripts/convert-org-to-typst.mjs:536-537</code> . A later pass should compare every surrounding Scheme result line, but the known placeholder bug appears repaired. |
| C3-010 | Chapter 1 footnote numbering | BUG            | The published page around the first Cartan discussion shows where <code>d/dt</code> is a vector field on the real line <sup>8</sup> . Source location: <code>fdg-book/scheme/org/chapter001.org:455</code> contains <code>[fn:8]</code> ; generated location: <code>typ/content/chapter001.typ:208-212</code> is the corresponding Typst footnote. Current output shows the marker as 14, because Typst footnotes are numbered continuously through front matter/prologue instead of following the published sequence for chapter 1. Repair entry point is likely a global footnote-numbering policy/helper, not the local paragraph. Published footnote numbering is now a fidelity requirement.            |
| C3-011 | Chapter 2 intro footnote     | PARTIAL-EX-ACT | The latitude/longitude footnote is present. Org source is <code>fdg-book/scheme/org/chapter002.org:368-371</code> , with <code>\$180^\circ\$</code> ; generated/current Typst is <code>typ/content/chapter002.typ:10</code> , with <code>180° meridian</code> . Converter repair                                                                                                                                                                                                                                                                                                                                                                                                                             |

|        |                               |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------|-------------------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        |                               |                | entry point for this known fix is <code>scripts/convert-org-to-typst.mjs:599</code> . Published extraction shows <code>180° meridian</code> ; glyph choice and spacing still need visual comparison, but content is present.                                                                                                                                                                                                                                                                                                                                                                                        |
| C3-012 | Chapter 2 section 2.3 repairs | PARTIAL-EX-ACT | Known source/PDF drift fixes in the coordinate-independence section are present. Org heading source is <code>fdg-book/scheme/org/chapter002.org:159</code> , with <code>Indepenedent</code> ; generated heading is <code>typ/content/chapter002.typ:102</code> , corrected to <code>Independent</code> . The repaired prose functions that map is at <code>typ/content/chapter002.typ:116</code> ; converter repair entry point includes <code>scripts/convert-org-to-typst.mjs:206</code> plus nearby chapter 2 targeted repairs around <code>:207-209</code> . Re-check this cluster when regenerating chapter 2. |
| C3-013 | Chapter 2 formulas            | PARTIAL-EX-ACT | Prior notes say cardioid formulas were repaired from PDF/source drift. Org source is <code>fdg-book/scheme/org/chapter002.org:256-277</code> , including the bad traditional formula at <code>:277</code> ; generated/current Typst is <code>typ/content/chapter002.typ:193-207</code> , now showing $r = 2a(1 + \cos(\theta))$ and $(x^2 + y^2 - 2ax)^2 = 4a^2(x^2 + y^2)$ . A later visual/equation pass should verify exact math styling, equation breaks, punctuation, and display/inline placement. Repair should remain converter-owned so regeneration preserves the PDF-derived fix.                        |
| C3-014 | Chapter 2 figures             | UNCHECKED      | Figures 2-1, 2-2, and 2-3 are included in Typst at <code>typ/content/chapter002.typ:37</code> , <code>:94</code> , and <code>:218</code> ; assets are <code>typ/assets/figures/fig-2-1.pdf</code> , <code>fig-2-2.pdf</code> , and <code>fig-2-3.pdf</code> ; helper is <code>typ/fdg-lib/figures.typ:3-11</code> . This chunk did not raster-compare figure placement, scaling, caption punctuation, or adjacency to the published PDF. Leave this as an explicit visual-audit target for the chapter 2 exhaustive pass.                                                                                           |
| C3-015 | Chunk 3 completion marker     | UNCHECKED      | This chunk still needs normalized section-level text diffs for every paragraph, footnote, equation, Scheme block, result block, exercise, figure caption, and cross-reference in preface, prologue, chapter 1, and chapter 2. Current entries are enough to guide the next exhaustive run, not enough to close the chunk as exact.                                                                                                                                                                                                                                                                                  |

Chunk 3 evidence commands:

```
pdftotext fdg-book/fdg_book.pdf /private/tmp/fdg-published-all.txt
pdftotext fdg-book.pdf /private/tmp/fdg-current-all.txt
rg -n "Papert|lagrangian|90|Cartan|coordinate independent|functions that map|cardioid|180|Stokes|
Acknowledgements|Galileo|Assayer" \
  fdg-book/scheme/org/prologue.org \
  fdg-book/scheme/org/chapter001.org \
  fdg-book/scheme/org/chapter002.org \
  typ/content/preface.typ \
  typ/content/prologue.typ \
  typ/content/chapter001.typ \
  typ/content/chapter002.typ \
  /private/tmp/fdg-published-all.txt \
  /private/tmp/fdg-current-all.txt
sed -n '492,530p' /private/tmp/fdg-published-all.txt
sed -n '368,390p' /private/tmp/fdg-current-all.txt
```

```
sed -n '748,770p' /private/tmp/fdg-published-all.txt
sed -n '554,566p' /private/tmp/fdg-current-all.txt
```

## Chunk 4 Audit Entries: Chapters 3-5

| ID     | Region                        | Status         | Finding                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------|-------------------------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C4-001 | Chunk scope                   | PARTIAL-EX-ACT | This chunk covers representative evidence from chapters 3, 4, and 5: published PDF text, Org source, generated Typst source, and current <code>fdg-book.pdf</code> extraction. It is not yet a complete paragraph/equation/code audit.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| C4-002 | Chapter 3 structure           | PARTIAL-EX-ACT | Chapter 3 is present as <b>Vector Fields and One-Form Fields</b> , with major sections including <b>Vector Fields</b> , <b>Coordinate-Basis Vector Fields</b> , integral curves, Lie derivatives, one-form fields, and exercises. Sampled opening prose and equation flow are present, but the chapter needs full equation/code/result comparison.                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| C4-003 | Chapter 3 literal hyphenation | BUG            | Visual check of current <code>fdg-book.pdf</code> physical page 28 confirms visible literal hyphenation artifacts. Exact locations: <code>fdg-book/scheme/org/chapter003.org:8</code> and generated <code>typ/content/chapter003.typ:6</code> contain <code>di- rection</code> ; <code>typ/content/chapter003.typ:23</code> contains <code>directional derivatives</code> ; <code>typ/content/chapter003.typ:35</code> contains <code>components</code> . Published PDF visual page for the chapter 3 opening shows normal line-break hyphenation without the extra post-hyphen space. Repair entry point: add a converter/post-merge cleanup in <code>scripts/convert-org-to-typst.mjs</code> near the existing targeted <code>.replaceAll(...)</code> repairs, then regenerate rather than hand-editing generated Typst. |
| C4-004 | Chapter 3 footnote numbering  | BUG            | Published chapter 3 starts its first note as footnote 1. Current output shows the first chapter 3 note as 25. Because published footnote numbering is now a fidelity requirement, chapter 3 and later chapters need a numbering policy/fix rather than treating this as layout drift.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| C4-005 | Chapter 3 equation references | BUG            | Visual/text checks confirm reference drift near the chapter 3 one-form discussion. Source location: <code>fdg-book/scheme/org/chapter003.org:628</code> says This follows from the dual relationship (3.41). Generated location: <code>typ/content/chapter003.typ:273</code> has <code>(@3.41)</code> , but current output resolves this as <code>((3.43))</code> ; published visual page 35 shows (3.41). Likely causes to inspect: equation labels around <code>typ/content/chapter003.typ:269-273</code> , equation creation in <code>scripts/convert-org-to-typst.mjs:136-146</code> , and reference rewriting in <code>scripts/convert-org-to-typst.mjs:1160-1196</code> . Fix should preserve the published reference number and avoid double parentheses.                                                           |
| C4-006 | Chapter 4 equation 4.3        | PARTIAL-EX-ACT | PDF-FIDELITY-NOTES.md records that chapter 4 equation 4.3 was repaired by restoring the missing $X_k$ subscript from the published PDF. Current Typst now renders the basis expansion as <code>sum_k sans(X)_k ... sans(c)_j^k</code> . Preserve this known repair and verify visually in the exhaustive pass.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| C4-007 | Equation 4.6 line break       | BUG            | Visual check clarifies that the useful issue here is equation layout, not the word <code>coefficents</code> . Published physical page 64 keeps equation (4.6) as one displayed equation: <code>sum_k d_k^i(m)c_j^k(m)</code> stays together before the equation num-                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

|        |                            |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------|----------------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        |                            |                | ber. Current physical page 42 splits the final factor $c_j^k(m)$ onto a new line below the equation number, making the display read awkwardly. Source locations: Org equation at <code>fdg-book/scheme/org/chapter004.org:80-83</code> ; generated Typst at <code>typ/content/chapter004.typ:42-45</code> . The prose <code>typo coefficients</code> appears in Org :86, generated Typst :47, and the published PDF, so preserve it unless a separate errata policy says to correct published typos. Repair entry point is the equation conversion/line-break handling for 4.6, not the following prose.                                    |
| C4-008 | Chapter 4 cross-references | BUG            | Published chapter 4 references property/equation (3.41) in the dual-basis introduction and before the <code>e-dual-basis</code> check. Source locations: <code>fdg-book/scheme/org/chapter004.org:30-31</code> , :39, and :135-136. Generated locations: <code>typ/content/chapter004.typ:16</code> , :20, and :79. Current visual page 41 resolves these as (3.43). Because equation/reference numbers should match the published PDF, fix chapter 3 equation labels first, then re-check all chapter 4 references to 3.40, 3.41, and nearby one-form equations.                                                                           |
| C4-009 | Figure 4.1                 | PARTIAL-EX-ACT | Visual check of current physical page 41 shows one visible figure 4.1 caption, not a visible duplicate caption. Source locations: Org placeholder/caption at <code>fdg-book/scheme/org/chapter004.org:42-49</code> , generated figure call at <code>typ/content/chapter004.typ:22</code> , and figure helper at <code>typ/fdg-lib/figures.typ:3-11</code> . <code>pdftotext</code> extraction shows both the generated Typst caption and embedded caption text from the imported figure PDF, so future automated diff tooling should ignore or normalize text embedded in figure assets. This is not currently a visible page-fidelity bug. |
| C4-010 | Figure 4.2                 | PARTIAL-EX-ACT | Visual check of current physical page 49 shows figure 4.2 and following prose are cleanly separated. Generated figure call: <code>typ/content/chapter004.typ:377</code> . The malformed <code>Take a point4.2 ...</code> sequence seen in text extraction is an extraction-order artifact, not a visible layout bug. Future text-based audits should cross-check suspect figure-adjacent extraction against rendered page images before filing layout bugs.                                                                                                                                                                                 |
| C4-011 | Chapter 4 exercises        | PARTIAL-EX-ACT | Exercises 4.2 and 4.3 are present in generated Typst. This chunk did not compare their full text against Org/published PDF, so keep them open for the exhaustive chapter 4 pass.                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| C4-012 | Chapter 5 structure        | PARTIAL-EX-ACT | Chapter 5 is present as <b>Integration</b> , with sections through <b>Stokes's Theorem</b> and associated exercises. Sampled opening integration formulas, wedge-product material, exterior derivative material, and Stokes material are present. Full equation/code/result comparison remains open.                                                                                                                                                                                                                                                                                                                                        |
| C4-013 | Figure 5.1                 | PARTIAL-EX-ACT | Visual check of current physical page 54 shows one visible figure 5.1 caption and no visible duplicate. Generated figure call: <code>typ/content/chapter005.typ:109</code> ; helper: <code>typ/fdg-lib/figures.typ:3-11</code> . Published physical page 80 has the corresponding single caption. <code>pdftotext</code> extraction duplicates/mangles caption text from the imported figure asset, e.g. <code>area 0of</code> ; classify this as text-layer/extraction behavior rather than a visible page-fidelity bug.                                                                                                                   |

|        |                              |                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------|------------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C4-014 | Chapter 5 equation numbering | <b>BUG</b>            | Visual checks confirm equation-number drift. Source: <code>fdg-book/scheme/org/chapter005.org:335-337</code> is the one-form exterior derivative display and <code>:342-348</code> is the general exterior derivative display. Generated: <code>typ/content/chapter005.typ:195-197</code> contains a displayed equation inside footnote 49 labelled <code>&lt;5.40&gt;</code> , then <code>:199-201</code> labels the one-form definition <code>&lt;5.23&gt;</code> , and <code>:203-207</code> labels the general definition <code>&lt;5.24&gt;</code> . Current visual page 55 shows footnote equation (5.23), pushing the visible body equations on page 56 to (5.24) and (5.25); published physical page 83 labels those body equations (5.23) and (5.24). Likely fix: prevent display equations inside footnotes from stepping the visible chapter equation counter, or manually set their published number/unnumbered behavior. |
| C4-015 | Equation 5.24 formula text   | <b>SOURCE-DRIFT</b>   | Visual checks confirm published equation (5.24) has the commutator term <code>omega([v_i, v_j], ...)</code> , while current equation (5.25) visibly shows <code>omega(v_i, v_j], ...)</code> with a closing bracket but no matching opening bracket. Source location: <code>fdg-book/scheme/org/chapter005.org:346</code> ; generated location: <code>typ/content/chapter005.typ:207</code> . Repair should be PDF-derived during the exhaustive chapter 5 equation pass. This likely requires a manual converter-side repair in <code>scripts/convert-org-to-typst.mjs</code> that both restores the missing opening commutator bracket and adds explicit alignment characters/line breaks so the display matches the published PDF layout, rather than relying on generic math conversion.                                                                                                                                          |
| C4-016 | Chapter 5 citation           | <b>PARTIAL-EX-ACT</b> | The Stokes theorem footnote cites Spivak, Calculus on Manifolds. Current reference numbering infrastructure should make this [17], but this chunk did not mechanically inspect the current footnote callout; verify during the full chapter 5 citation/footnote pass.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| C4-017 | Chunk 4 completion marker    | <b>UNCHECKED</b>      | Still unchecked for chapters 3-5: every equation number/reference target, all footnote numbers/text, all Scheme blocks and result blocks, all exercises, all figure visual crops/captions, and normalized paragraph text. Current entries identify high-value defects to fix before a fully exhaustive comparison.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

Chunk 4 evidence commands:

```
rg -n "di- rection|deriva- tives|com- ponents|3\\.41|3\\.43|Figure 4\\.1|Figure 4\\.2|Figure 5\\.1|5\\.24|coefficients" \
  fdg-book/scheme/org/chapter003.org \
  fdg-book/scheme/org/chapter004.org \
  fdg-book/scheme/org/chapter005.org \
  typ/content/chapter003.typ \
  typ/content/chapter004.typ \
  typ/content/chapter005.typ \
  /private/tmp/fdg-published-all.txt \
  /private/tmp/fdg-current-all.txt
sed -n '1568,1630p' /private/tmp/fdg-published-all.txt
sed -n '1518,1624p' /private/tmp/fdg-current-all.txt
sed -n '2492,2588p' /private/tmp/fdg-published-all.txt
sed -n '2450,2555p' /private/tmp/fdg-current-all.txt
sed -n '3548,3558p' /private/tmp/fdg-published-all.txt
```

```

sed -n '3518,3530p' /private/tmp/fdg-current-all.txt
sed -n '3698,3714p' /private/tmp/fdg-published-all.txt
sed -n '3748,3762p' /private/tmp/fdg-current-all.txt
sed -n '1,180p' typ/fdg-lib/figures.typ
pdftoppm -png -r 180 -f 28 -l 28 fdg-book.pdf /private/tmp/fdg-current-p28
pdftoppm -png -r 180 -f 41 -l 41 fdg-book.pdf /private/tmp/fdg-current-p41
pdftoppm -png -r 180 -f 49 -l 49 fdg-book.pdf /private/tmp/fdg-current-p49
pdftoppm -png -r 180 -f 54 -l 54 fdg-book.pdf /private/tmp/fdg-current-p54
pdftoppm -png -r 180 -f 56 -l 56 fdg-book.pdf /private/tmp/fdg-current-p56
pdftoppm -png -r 180 -f 28 -l 28 fdg-book/fdg_book.pdf /private/tmp/fdg-published-p28
pdftoppm -png -r 180 -f 42 -l 42 fdg-book/fdg_book.pdf /private/tmp/fdg-published-p42
pdftoppm -png -r 180 -f 80 -l 80 fdg-book/fdg_book.pdf /private/tmp/fdg-published-p80
pdftoppm -png -r 180 -f 83 -l 83 fdg-book/fdg_book.pdf /private/tmp/fdg-published-p83

```

## Chunk 5 Audit Entries: Chapters 6-8

| ID     | Region                | Status         | Finding                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------|-----------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C5-001 | Chunk scope           | PARTIAL-EX-ACT | This chunk covers representative evidence from chapters 6, 7, and 8: Org source, generated Typst, published PDF text extraction, current PDF extraction, and visual page checks for figure pages and the main chapter 8 numbering defect. It is not yet a full paragraph/equation/code/result audit.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| C5-002 | Chapter 6 structure   | PARTIAL-EX-ACT | Chapter 6 is present as <i>Over a Map</i> , with sections from vector fields over a map through pushforward along integral curves and exercises. Primary generated span begins at <code>typ/content/chapter006.typ:5</code> ; sampled Org source begins at <code>fdg-book/scheme/org/chapter006.org:1</code> ; published chapter 6 begins around published physical page 92. Full equation/code/result comparison remains open.                                                                                                                                                                                                                                                                                                                                                       |
| C5-003 | Figure 6.1            | PARTIAL-EX-ACT | Visual check of current physical page 62 shows one visible figure 6.1 caption, no visible duplicate caption, and a clean relation between the figure, caption, equations (6.1)-(6.2), and following code block. Generated figure call: <code>typ/content/chapter006.typ:14</code> ; asset: <code>typ/assets/figures/fig-6-1.pdf</code> ; helper: <code>typ/fdg-lib/figures.typ:3-11</code> . Published extraction places the matching figure/caption on published physical page 93. <code>pdftotext</code> reports a duplicate caption because embedded figure-asset text is extracted as well as the generated Typst caption; treat that as a text-layer artifact unless a rendered-page check later contradicts it.                                                                 |
| C5-004 | Chapter 6 Scheme typo | SOURCE-DRIFT   | The <code>form-field-&gt;form-field-over-map</code> listing in Org and current Typst calls ( <code>make fake-vector-field V-over-mu n</code> ), while the surrounding definition/prose and the published PDF use <code>make-fake-vector-field</code> . Source locations: <code>fdg-book/scheme/org/chapter006.org:117-131</code> , especially <code>:128</code> ; generated Typst: <code>typ/content/chapter006.typ:65-80</code> , especially <code>:76</code> ; current visual page 63 shows the spaced call; published physical page 95 shows <code>make-fake-vector-field</code> . Repair should be <code>converter/source-aware</code> : either patch upstream Org input or add a targeted converter repair in <code>scripts/convert-org-to-typst.mjs</code> before regeneration. |
| C5-005 | Chapter 6 Scheme typo | SOURCE-DRIFT   | The sphere coordinate-system setup in Org and current Typst says ( <code>coordinate-system at 'spherical 'north-pole S2</code> ), but the published PDF uses ( <code>coordinate-system-at 'spherical 'north-pole S2</code> ). Source locations: <code>fdg-</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|        |                               |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------|-------------------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        |                               |                | book/scheme/org/chapter006.org:168-174, especially :171; generated Typst: typ/content/chapter006.typ:100-106, especially :103; current visual page 64 shows the spaced form; published physical page 96 shows the hyphenated procedure name. Repair path is the same as C5-004: source correction or a targeted converter repair, then regenerate.                                                                                                                                                                                                                                                                                            |
| C5-006 | Chapter 6 pullback code       | SOURCE-DRIFT   | The pullback function listing in Org and current Typst has a variable-name mismatch: parameter f-on-m but body f-on-M. Published PDF uses f-on-M consistently. Source locations: fdg-book/scheme/org/chapter006.org:275-279; generated Typst: typ/content/chapter006.typ:181-186; current physical page 65 shows f-on-m followed by f-on-M; published physical page 98 shows (define ((pullback-function mu:N->M) f-on-M) and (compose f-on-M mu:N->M)). Treat as a source/published drift needing a targeted source or converter repair, not a layout issue.                                                                                 |
| C5-007 | Chapter 7 opening typo        | SOURCE-DRIFT   | Org and current Typst say We can asume without loss of generality; published PDF says We can assume without loss of generality. Source locations: fdg-book/scheme/org/chapter007.org:38-41; generated Typst: typ/content/chapter007.typ:16; current visual/text page 69 shows asume; published physical page 104 shows assume. This is a good early source-drift repair target because it is visible prose, not typography.                                                                                                                                                                                                                   |
| C5-008 | Equation 7.9 repair           | PARTIAL-EX-ACT | PDF-FIDELITY-NOTES.md records the existing repair for chapter 7 equation 7.9: punctuation was moved out of $\mathsf{g}$ . Current generated Typst at typ/content/chapter007.typ:57 renders the final factor as sans(g) followed by punctuation, rather than including the period inside the styled g. Org source location is fdg-book/scheme/org/chapter007.org:101-103, where the original $\mathsf{g}$ defect appears. Preserve this known PDF-derived repair and visually re-check during the exhaustive chapter 7 equation pass.                                                                                                          |
| C5-009 | Equation 7.14 notation/layout | PARTIAL-EX-ACT | Generated Typst for equation 7.14 contains $\phi_{\text{alt}(\delta)}^{(\text{sans}(v))}$ at typ/content/chapter007.typ:102. Org source fdg-book/scheme/org/chapter007.org:176-183 has $\phi_{\mathsf{v}}^{\delta}$ ; published physical page 107 visually shows the intended pushforward star attached to the lower delta, not a literal delta * word sequence. Current visual page 71 renders the display as a mathematical subscript-star expression, so this is not yet filed as a visible formula bug. However, the Typst source is fragile and should be revisited if equation 7.14 is manually aligned or normalized by the converter. |
| C5-010 | Figure 7.1                    | PARTIAL-EX-ACT | Visual check of current physical page 82 shows one visible figure 7.1 caption, a clean figure crop, and no visible duplicate caption. Generated figure call: typ/content/chapter007.typ:654; asset: typ/assets/figures/fig-7-1.pdf; helper: typ/fdg-lib/figures.typ:3-11. Published extraction places the corresponding caption on published physical page 125. As with other imported figures, pdftotext sees both generated caption text and embedded asset text; future automated audits should normalize figure                                                                                                                           |

|        |                                |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------|--------------------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        |                                |                | text layers or require image-page confirmation before filing caption duplication.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| C5-011 | Chapter 8 structure            | PARTIAL-EX-ACT | Chapter 8 is present as <i>Curvature</i> , with opening Riemann curvature definitions, explicit transport, verification, torsion, and Bianchi identities. Generated span begins at <code>typ/content/chapter008.typ:5</code> ; sampled Org source begins at <code>fdg-book/scheme/org/chapter008.org:1</code> ; current chapter starts on current physical page 90; published chapter 8 starts around published physical page 136. Full equation/code/result comparison remains open.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| C5-012 | Chapter 8 equation numbering   | BUG            | Visual check confirms a body equation-numbering drift caused by numbered display equations inside a footnote. Generated Typst puts footnote equations <code>&lt;8.33&gt;</code> and <code>&lt;8.34&gt;</code> inside the footnote at <code>typ/content/chapter008.typ:91-100</code> , then labels the next body displays <code>&lt;8.10&gt;</code> , <code>&lt;8.11&gt;</code> , and <code>&lt;8.12&gt;</code> at <code>:102-115</code> . Current visual page 92 instead prints those body equations as (8.12), (8.13), and (8.14), while the footnote prints (8.10) and (8.11). Published physical page 139 prints the body equations as (8.10), (8.11), and (8.12). Likely fix is shared with C4-014: display equations inside footnotes should not advance the visible body equation counter, or they need explicit published-number handling. Inspect equation creation/counter behavior in <code>typ/fdg-lib/chapter.typ</code> and equation conversion in <code>scripts/convert-org-to-typst.mjs</code> . |
| C5-013 | Chapter 8 punctuation          | SOURCE-DRIFT   | Org and current Typst contain a visible space before comma in <i>However</i> , if ...; the published PDF has <i>However</i> , if .... Source location: <code>fdg-book/scheme/org/chapter008.org:166-169</code> ; generated Typst: <code>typ/content/chapter008.typ:113</code> ; current visual page 92 shows <i>However</i> ,; published physical page 139 shows <i>However</i> ,. This can be corrected either in source or by a narrow converter cleanup for space-before-punctuation artifacts.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| C5-014 | Chapter 8 symmetric connection | SOURCE-DRIFT   | Org and current Typst state the symmetric-connection condition tautologically as $\Gamma_{(j\ k)}^i = \Gamma_{(j\ k)}^i$ . Published PDF correctly shows $\Gamma_{jk}^i = \Gamma_{kj}^i$ . Source location: <code>fdg-book/scheme/org/chapter008.org:743-744</code> ; generated Typst: <code>typ/content/chapter008.typ:553-556</code> ; current visual page 100 shows identical lower indices on both sides; published physical page 150 shows the lower indices swapped on the right. This should be a PDF-derived mathematical repair in source/converter output, not a style adjustment.                                                                                                                                                                                                                                                                                                                                                                                                                    |
| C5-015 | Chapter 8 torsion errata       | PROJECT-ADDED  | Current end-matter errata notes that the torsion example on page 127 uses an undefined <i>f</i> . That errata is intentional project-added material, so do not treat it as a published-body mismatch. During the exhaustive chapter 8 pass, compare the body example against published PDF and compare the errata note against <code>fdg-book/scheme/org/errata.org / typ/content/errata.typ</code> separately.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| C5-016 | Chunk 5 completion marker      | UNCHECKED      | Still unchecked for chapters 6-8: every equation number/reference target, all published footnote numbers/text, every Scheme block and result block, all exercises, all citations, all section-title line breaks, and figure scaling against the                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

|  |  |  |                                                                                                                                                       |
|--|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  |  | published pages. Current entries identify high-value source drift and one shared equation-counter defect to fix before a fully exhaustive comparison. |
|--|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------|

Chunk 5 evidence commands:

```
nl -ba typ/content/chapter006.typ | sed -n '1,220p'
nl -ba typ/content/chapter007.typ | sed -n '1,220p;640,670p;948,1002p'
nl -ba typ/content/chapter008.typ | sed -n '1,240p;548,584p'
nl -ba fdg-book/scheme/org/chapter006.org | sed -n '70,210p;220,310p'
nl -ba fdg-book/scheme/org/chapter007.org | sed -n '1,210p'
nl -ba fdg-book/scheme/org/chapter008.org | sed -n '1,190p;720,770p'
pdftotext -f 62 -l 65 fdg-book.pdf -
pdftotext -f 93 -l 98 fdg-book/fdg_book.pdf -
pdftotext -f 69 -l 71 fdg-book.pdf -
pdftotext -f 104 -l 107 fdg-book/fdg_book.pdf -
pdftotext -f 82 -l 82 fdg-book.pdf -
pdftotext -f 125 -l 125 fdg-book/fdg_book.pdf -
pdftotext -f 92 -l 92 fdg-book.pdf -
pdftotext -f 139 -l 139 fdg-book/fdg_book.pdf -
pdftotext -f 100 -l 101 fdg-book.pdf -
pdftotext -f 150 -l 151 fdg-book/fdg_book.pdf -
pdftoppm -f 62 -singlefile -png -r 130 fdg-book.pdf /tmp/fdg-current-p62
pdftoppm -f 82 -singlefile -png -r 130 fdg-book.pdf /tmp/fdg-current-p82
pdftoppm -f 92 -singlefile -png -r 130 fdg-book.pdf /tmp/fdg-current-p92
```

## Chunk 6 Audit Entries: Chapters 9-11

| ID     | Region                            | Status         | Finding                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------|-----------------------------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C6-001 | Chunk scope                       | PARTIAL-EX-ACT | This chunk covers representative evidence from chapters 9, 10, and 11: Org source, generated Typst, published PDF text extraction, current PDF extraction, and rendered-page checks for chapter 10 formulas and chapter 11 reconstructed equations. It is not yet a complete prose/equation/code/result audit.                                                                                                                                                                                                                                                                                         |
| C6-002 | Chapter 9 structure               | PARTIAL-EX-ACT | Chapter 9 is present as <b>Metrics</b> , with opening metric examples, <b>Metric Music</b> , metric compatibility, Lagrange/geodesic discussion, Newtonian approximation, and Einstein-equation material. Generated span begins at <code>typ/content/chapter009.typ:5</code> ; sampled Org source begins at <code>fdg-book/scheme/org/chapter009.org:1</code> ; current chapter 9 starts on current physical page 103; published chapter 9 starts around published physical page 154. Full equation/code/result comparison remains open.                                                               |
| C6-003 | Chapter 9 cross-reference wording | PARTIAL-EX-ACT | Exercise 9.1 refers to the literal metric discussion. Org/published prose says <b>as described on page 6</b> ; current generated Typst at <code>typ/content/chapter009.typ:156-158</code> uses <b>Section #fdg-ref-page(&lt;sec-1.2&gt;)</b> , which renders on current physical page 106 as <b>Section 1.2, page 3</b> . Published physical page 158 says <b>on page 6</b> . Because page geometry is intentionally different, the page number itself is accepted; the extra <b>Section 1.2</b> wording is a generated-reference style difference to decide during the full chapter 9 reference pass. |
| C6-004 | Chapter 9 equations               | PARTIAL-EX-ACT | Sampled current output preserves equation numbering through the metric-compatibility span: current physical pages 105-107 show equations (9.11)-(9.20), and published physical pages 157-160 show the matching numbers. This                                                                                                                                                                                                                                                                                                                                                                           |

|        |                            |                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------|----------------------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        |                            |                    | chunk did not visually compare every symbol inside those equations, so keep all chapter 9 equations open for the exhaustive pass.                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| C6-005 | Chapter 10 structure       | PARTIAL-EX-ACT     | Chapter 10 is present as <b>Hodge Star and Electrodynamics</b> , with Hodge dual definition, vector-calculus relationship, spherical coordinates, wave equation, electrodynamics, Lorentz force, and exercise 10.1. Generated span begins at <code>typ/content/chapter010.typ:5</code> ; sampled Org source begins at <code>fdg-book/scheme/org/chapter010.org:1</code> ; current chapter 10 starts on current physical page 117; published chapter 10 starts on published physical page 174. Full formula/code/result comparison remains open.                             |
| C6-006 | Chapter 10 source repairs  | PDF-DERIVED-REPAIR | Two early chapter 10 source defects are already repaired by the converter. Org <code>fdg-book/scheme/org/chapter010.org:44-46</code> contains malformed $g^{\{ij\}}$ and an unclosed epsilon subscript; converter repairs are in <code>scripts/convert-org-to-typst.mjs:226-227</code> ; generated Typst renders the corrected Hodge-dual prose at <code>typ/content/chapter010.typ:16</code> . Preserve these repairs and visually re-check in the full chapter 10 opening-equation pass.                                                                                  |
| C6-007 | Equation 10.3 formula      | SOURCE-DRIFT       | Visual check confirms a visible formula mismatch. Org source <code>fdg-book/scheme/org/chapter010.org:81-84</code> and generated Typst <code>typ/content/chapter010.typ:31-33</code> render equation (10.3) as $\text{grad}(f) = g^{\text{sharp}}(df)(df)$ . Current visual page 118 shows the trailing $(df)$ . Published physical page 175 shows $\text{grad}(f) = g^{\text{sharp}}(df)$ with no trailing argument. Repair should be PDF-derived, likely a targeted converter/source repair for this equation rather than a generic math rewrite.                         |
| C6-008 | Equations 10.5-10.6 layout | BUG                | Visual check of current pages 118-119 shows equations (10.5) and (10.6) broken into narrow vertical stacks, making the formulas much harder to read than the published layout. Published physical page 176 keeps equation (10.5) compact across two lines and equation (10.6) compact across three lines. Source locations: <code>fdg-book/scheme/org/chapter010.org:101-124</code> ; generated Typst: <code>typ/content/chapter010.typ:46-62</code> . This likely needs manual converter-side alignment/line-break repair for these displays, similar in spirit to C4-015. |
| C6-009 | Equation 10.6 formula      | SOURCE-DRIFT       | Visual/source check confirms a semantic formula mismatch inside equation (10.6). Org <code>fdg-book/scheme/org/chapter010.org:115-124</code> and generated Typst <code>typ/content/chapter010.typ:57-62</code> put the second curl component on $dx$ ; current visual page 119 shows the second basis form as $dx$ . Published physical page 176 shows that second component on $dy$ , followed by the third on $dz$ . Repair should be PDF-derived and probably bundled with the equation 10.6 alignment repair.                                                           |
| C6-010 | Chapter 11 structure       | PARTIAL-EX-ACT     | Chapter 11 is present as <b>Special Relativity</b> , with opening electromagnetism motivation, wave-equation invariance, Lorentz transformations, implementation, rotations, general Lorentz transformations, and exercises. Generated span begins at <code>typ/content/chapter011.typ:5</code> ; sampled Org source begins at <code>fdg-book/scheme/org/chapter011.org:1</code> ; current                                                                                                                                                                                  |

|        |                                    |                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------|------------------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        |                                    |                    | chapter 11 starts on current physical page 130; published chapter 11 starts on published physical page 188. Full comparison remains open.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| C6-011 | Chapter 11 reconstructed equations | PDF-DERIVED-REPAIR | PDF-FIDELITY-NOTES.md:25-29 records that Org contains empty equation environments for equations 11.1-11.14 and 11.20-11.21, so the converter reconstructs them from the published PDF. Org empty-equation span starts at fdg-book/scheme/org/chapter011.org:48-119 and continues for 11.12-11.14 at :154-168, then 11.20-11.21 at :200-208. Converter insertion table is scripts/convert-org-to-typst.mjs:687-754; generated equations are visible at typ/content/chapter011.typ:16-62 and :73-105. Visual checks: current pages 130-132 show equations (11.1)-(11.14) and (11.20)-(11.21); published pages 189-192 show the matching published sequence. Keep these as intentional PDF-derived repairs, not Org/Typst drift. |
| C6-012 | Chapter 11 typo repair             | PDF-DERIVED-REPAIR | Org and the published PDF contain Einstein in the opening chapter 11 discussion: source fdg-book/scheme/org/chapter011.org:24; published physical page 188. The converter intentionally repairs this via scripts/convert-org-to-typst.mjs:229, and current visual page 130 shows Einstein. This is an intentional project/PDF-derived correction recorded in prior notes, not a regression.                                                                                                                                                                                                                                                                                                                                   |
| C6-013 | Chapter 11 Lorentz interval        | PDF-DERIVED-REPAIR | Org source fdg-book/scheme/org/chapter011.org:224-226 has the interval typo $-(x_i^0)^2 + (x_i^2)^2 + (x_i^2)^2 + (x_i^3)^2$ ; generated Typst typ/content/chapter011.typ:112 renders the repaired $(x_i^1)^2$ term after converter cleanup at scripts/convert-org-to-typst.mjs:239. Preserve this repair and visually verify against the published equation (11.22) during the full chapter 11 equation pass.                                                                                                                                                                                                                                                                                                                |
| C6-014 | Chapter 11 double reference        | BUG                | Current physical page 131 visibly renders the Lorentz-force cross-reference as ((11.11)). Generated Typst source typ/content/chapter011.typ:68 contains (@11.11), inherited from Org fdg-book/scheme/org/chapter011.org:140, where the reference was literal prose (11.11). Published physical page 191 shows a single parenthesized reference. Likely repair entry point is the same reference-rewriting/double-parentheses path noted in C4-005: inspect scripts/convert-org-to-typst.mjs:1160-1196 and normalize already-parenthesized equation references.                                                                                                                                                                |
| C6-015 | Chapter 11 code repairs            | PDF-DERIVED-REPAIR | Prior notes record converter repairs for make-4-tuple, $R^{\{=1\}}$ , and $\bar{v}$ .double norm notation. Concrete anchors: Org fdg-book/scheme/org/chapter011.org:461 has (make-4-tuple; generated typ/content/chapter011.typ:253 uses (make-4tuple; Org :524 has $R^{\{=1\}}$ ; generated typ/content/chapter011.typ:303 uses $R^{\{-1\}}$ . Converter cleanup locations include scripts/convert-org-to-typst.mjs:241-242 and :766-768. Preserve these repairs; full chapter 11 code/result audit remains open.                                                                                                                                                                                                            |
| C6-016 | Chunk 6 completion marker          | UNCHECKED          | Still unchecked for chapters 9-11: every equation symbol and alignment, all code/result blocks, all footnote numbers/text, all citations, every cross-reference, and all exercises. Highest-priority follow-up items from this chunk are equation 10.3,                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

|  |  |  |                                                                                              |
|--|--|--|----------------------------------------------------------------------------------------------|
|  |  |  | equation 10.6, chapter 10 display alignment, and chapter 11 double-parenthesized references. |
|--|--|--|----------------------------------------------------------------------------------------------|

Chunk 6 evidence commands:

```
nl -ba typ/content/chapter009.typ | sed -n '1,240p'
nl -ba typ/content/chapter010.typ | sed -n '1,280p'
nl -ba typ/content/chapter011.typ | sed -n '1,310p'
nl -ba fdg-book/scheme/org/chapter009.org | sed -n '1,260p'
nl -ba fdg-book/scheme/org/chapter010.org | sed -n '1,280p'
nl -ba fdg-book/scheme/org/chapter011.org | sed -n '1,280p;420,545p'
nl -ba scripts/convert-org-to-typst.mjs | sed -n '220,250p;680,772p'
pdftotext -f 105 -l 107 fdg-book.pdf -
pdftotext -f 157 -l 160 fdg-book/fdg_book.pdf -
pdftotext -f 117 -l 119 fdg-book.pdf -
pdftotext -f 174 -l 176 fdg-book/fdg_book.pdf -
pdftotext -f 130 -l 137 fdg-book.pdf -
pdftotext -f 188 -l 192 fdg-book/fdg_book.pdf -
pdftoppm -f 118 -singlefile -png -r 130 fdg-book.pdf /tmp/fdg-current-p118
pdftoppm -f 119 -singlefile -png -r 130 fdg-book.pdf /tmp/fdg-current-p119
pdftoppm -f 175 -singlefile -png -r 130 fdg-book/fdg_book.pdf /tmp/fdg-published-p175
pdftoppm -f 176 -singlefile -png -r 130 fdg-book/fdg_book.pdf /tmp/fdg-published-p176
pdftoppm -f 132 -singlefile -png -r 130 fdg-book.pdf /tmp/fdg-current-p132
pdftoppm -f 192 -singlefile -png -r 130 fdg-book/fdg_book.pdf /tmp/fdg-published-p192
```

## Chunk 7 Audit Entries: Appendices A-C

| ID     | Region                      | Status         | Finding                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------|-----------------------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C7-001 | Chunk scope                 | PARTIAL-EX-ACT | This chunk covers representative evidence from appendices A, B, and C: Org source, generated Typst, published/current text extraction, and rendered-page spot checks around Appendix B selector/product equations and Appendix C tensor equations. It is not yet a full code/result/equation audit.                                                                                                                                                                                 |
| C7-002 | Appendix A structure        | PARTIAL-EX-ACT | Appendix A is present as Scheme, with the IEEE Scheme epigraph, procedure calls, lambda expressions, definitions, conditionals, recursion, local names, compound data, and symbols. Generated span begins at typ/content/appendix_a.typ:5; sampled Org source begins at fdg-book/scheme/org/appendix_a.org:1; current appendix A begins on current physical page 143; published appendix A begins on published physical page 206. Full Scheme block/result comparison remains open. |
| C7-003 | Appendix A references       | PARTIAL-EX-ACT | The Appendix A epigraph and opening prose preserve the intended references after bibliography-order seeding: Org fdg-book/scheme/org/appendix_a.org:12-17 has [10] and [1]; generated Typst typ/content/appendix_a.typ:6-12 cites @ieee1991scheme and @abelson1996sicp; current physical page 143 shows IEEE Scheme [10] and SICP [1]. Published physical page 206 shows the same reference numbers. Styling of the block quote remains open.                                       |
| C7-004 | Appendix footnote numbering | BUG            | Published appendix A starts its first appendix footnote as 1; current physical page 143 shows the first appendix A footnote as 108. This is the same global footnote-numbering fidelity problem recorded earlier, but it should be tracked for appendices too. Generated first appendix A note is at typ/content/appendix_a.typ:12; source note is fdg-book/                                                                                                                        |

|        |                             |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------|-----------------------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        |                             |                | scheme/org/appendix_a.org:15. Fix should come from the overall published-footnote-numbering policy rather than local appendix edits.                                                                                                                                                                                                                                                                                                                                                                  |
| C7-005 | Appendix B structure        | PARTIAL-EX-ACT | Appendix B is present as <b>Our Notation</b> , with functional notation, generic arithmetic, functions, symbolic values, tuples, derivatives, structured results, and exercises. Generated span begins at typ/content/appendix_b.typ:5; sampled Org source begins at fdg-book/scheme/org/appendix_b.org:1; current appendix B begins on current physical page 149; published appendix B begins around published physical page 216. Full equation/code/result comparison remains open.                 |
| C7-006 | Equation B.7 selector       | SOURCE-DRIFT   | Visual/text checks confirm a visible mathematical mismatch. The phase-space tuple is $s = (t, (x, y), [p_x, p_y])$ , so published physical page 220 has $I_0(s) = t$ . Org source fdg-book/scheme/org/appendix_b.org:244-253 and generated Typst typ/content/appendix_b.typ:143-151 instead have $I_0(s) = y$ ; current physical page 152 shows $I_0(s) = y$ . Repair should be PDF-derived, likely a targeted source/converter correction for equation B.7.                                          |
| C7-007 | Equation B.10 product       | SOURCE-DRIFT   | Visual/text checks confirm another Appendix B source drift. The prose says If A and C are not compatible for contraction and $C = [C_0, C_1, C_2]$ ; published physical page 222 labels the product as $AC = [AC_0, AC_1, AC_2]$ . Org source fdg-book/scheme/org/appendix_b.org:338-343 and generated Typst typ/content/appendix_b.typ:199-201 instead label the left side A B; current physical page 153 shows $AB = [AC_0, AC_1, AC_2]$ . Repair should be PDF-derived and local to equation B.10. |
| C7-008 | Appendix B matrix displays  | PARTIAL-EX-ACT | Current physical page 153 shows equations B.11-B.14 in a compact but readable letter-layout form. Published physical page 222 uses the published smaller page geometry and different line breaks. This chunk did not verify every symbol in B.11-B.14, so leave the matrix/tuple display sequence open for the exhaustive Appendix B equation pass.                                                                                                                                                   |
| C7-009 | Appendix C structure        | PARTIAL-EX-ACT | Appendix C is present as <b>Tensors</b> , with tensor definition, basis-change formulas, Riemann tensor test, torsion test, and the covariant-derivative non-tensor example. Generated span begins at typ/content/appendix_c.typ:5; sampled Org source begins at fdg-book/scheme/org/appendix_c.org:1; current appendix C begins on current physical page 159; published appendix C begins on published physical page 232. Full tensor-equation/code/result comparison remains open.                  |
| C7-010 | Equation C.1 multilinearity | SOURCE-DRIFT   | Visual check confirms a mathematical source drift in the first tensor-linearity equation. Published physical page 232 has $T(fu + gv, \omega) = f T(u, \omega) + g T(v, \omega)$ . Org source fdg-book/scheme/org/appendix_c.org:19-22 and generated Typst typ/content/appendix_c.typ:8-14 render the second term as $g T(u, \omega)$ ; current physical page 159 shows the same incorrect repeated u. Repair should be PDF-derived and local to equation C.1.                                        |

|        |                            |                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------|----------------------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C7-011 | Equation transform C.10    | <b>SOURCE-DRIFT</b>   | Visual check confirms a tensor transformation mismatch. Published physical page 234 has the primed Riemann component on the left side of equation C.10. Org source <code>fdg-book/scheme/org/appendix_c.org:129-133</code> and generated Typst <code>typ/content/appendix_c.typ:79-81</code> put the unprimed component on the left. Current physical page 160 shows the unprimed left side. Repair should be PDF-derived and coordinated with C.8-C.10 notation so prime placement is consistent across the transformation pair. |
| C7-012 | Appendix C cross-reference | <b>BUG</b>            | Current Appendix C says torsion is <b>see equation (8.23)</b> because chapter 8 equation numbering is already drifted by footnote display equations. Generated source <code>typ/content/appendix_c.typ:95</code> references <code>@8.21</code> ; published physical page 235 says <b>see equation 8.21</b> . This should be rechecked after C5-012 fixes chapter 8 footnote-equation counter behavior. Do not patch Appendix C locally unless the chapter 8 numbering fix leaves this reference wrong.                            |
| C7-013 | Appendix C code/results    | <b>PARTIAL-EX-ACT</b> | The sampled tensor-test and torsion-test code blocks are present in current output. Current physical pages 160-161 show <code>tensor-test, ;; (0 0 0 0)</code> , torsion test, and <code>;; (0 0 MESS)</code> ; published physical pages 234-236 show the corresponding code/results, though extraction differs in comment prefixes and code styling. Full code/result comparison remains open.                                                                                                                                   |
| C7-014 | Chunk 7 completion marker  | <b>UNCHECKED</b>      | Still unchecked for appendices A-C: every Scheme block/result pair, all footnote text and numbering, every equation symbol/alignment, all appendix citations, and all appendix cross-references back into the main text. Highest-priority follow-up items from this chunk are B.7, B.10, C.1, C.10, and the chapter 8 reference drift visible in Appendix C.                                                                                                                                                                      |

Chunk 7 evidence commands:

```
nl -ba typ/content/appendix_a.typ | sed -n '1,260p'
nl -ba typ/content/appendix_b.typ | sed -n '1,760p'
nl -ba typ/content/appendix_c.typ | sed -n '1,260p'
nl -ba fdg-book/scheme/org/appendix_a.org | sed -n '1,260p'
nl -ba fdg-book/scheme/org/appendix_b.org | sed -n '1,760p'
nl -ba fdg-book/scheme/org/appendix_c.org | sed -n '1,260p'
pdftotext -f 143 -l 149 fdg-book.pdf -
pdftotext -f 206 -l 216 fdg-book/fdg_book.pdf -
pdftotext -f 152 -l 153 fdg-book.pdf -
pdftotext -f 220 -l 222 fdg-book/fdg_book.pdf -
pdftotext -f 159 -l 161 fdg-book.pdf -
pdftotext -f 232 -l 236 fdg-book/fdg_book.pdf -
pdftoppm -f 152 -singlefile -png -r 130 fdg-book.pdf /tmp/fdg-current-p152
pdftoppm -f 153 -singlefile -png -r 130 fdg-book.pdf /tmp/fdg-current-p153
pdftoppm -f 220 -singlefile -png -r 130 fdg-book/fdg_book.pdf /tmp/fdg-published-p220
pdftoppm -f 222 -singlefile -png -r 130 fdg-book/fdg_book.pdf /tmp/fdg-published-p222
pdftoppm -f 159 -singlefile -png -r 130 fdg-book.pdf /tmp/fdg-current-p159
pdftoppm -f 160 -singlefile -png -r 130 fdg-book.pdf /tmp/fdg-current-p160
pdftoppm -f 232 -singlefile -png -r 130 fdg-book/fdg_book.pdf /tmp/fdg-published-p232
pdftoppm -f 234 -singlefile -png -r 130 fdg-book/fdg_book.pdf /tmp/fdg-published-p234
```

## Chunk 8 Audit Entries: Index and Errata

| ID | Region | Status | Finding |
|----|--------|--------|---------|
|----|--------|--------|---------|

|        |                               |                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------|-------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C8-001 | Chunk scope                   | <b>PARTIAL-EX-ACT</b>     | This chunk covers end-matter policy and representative evidence for the published index, the generated Typst index, and project errata. It uses line-level source anchors, text extraction, and visual checks of current physical pages 163 and 166 plus published physical page 240. It is not an exhaustive term-by-term index audit and does not implement any fixes.                                                                                                                                                                                                                                                                                                                                                                                    |
| C8-002 | Index comparison policy       | <b>ACCEPTED-STYLISTIC</b> | Index page locators cannot be required to match exactly because the current book deliberately uses US-letter geometry, omits the published cover/index page context, and has different body pagination. Future index audit should compare entry vocabulary, grouping/subentry structure, ordering, symbol/code styling, footnote locator notation, and procedure-definition locator styling. Numeric locators should be checked only for local consistency within each artifact unless a future build intentionally reproduces published pagination.                                                                                                                                                                                                        |
| C8-003 | Index generation architecture | <b>PARTIAL-EX-ACT</b>     | The current index is generated by <code>typ/index.typ</code> : <code>prose-trigger show rules span typ/index.typ:13-259</code> , <code>Scheme/code-trigger rules span typ/index.typ:261-367</code> , and the rendered index page is produced by <code>fdg-index-page</code> at <code>typ/index.typ:372-390</code> . The converter imports these helpers at <code>scripts/convert-org-to-typst.mjs:1295-1297</code> , wraps all non-errata content in <code>#fdg-indexed-body[...]</code> at <code>:1303-1305</code> , emits <code>#fdg-index-page()</code> at <code>:1306-1307</code> , then includes errata at <code>:1308</code> . This means index entries are generated from pattern matches in the rendered body, not from the published index source. |
| C8-004 | Index start page              | <b>PARTIAL-EX-ACT</b>     | Visual check: current physical page 163 starts the generated <b>Index</b> after current references; published physical page 240 starts the published <b>Index</b> . Both begin with the Knuth quotation and then index columns. Current extraction confirms pages 163-165 contain the current index; published extraction confirms pages 240-249 contain the published index. Full end-page verification of every published index page remains open.                                                                                                                                                                                                                                                                                                        |
| C8-005 | Index explanatory note        | <b>BUG</b>                | The explanatory note does not match the published policy. Published physical page 240 visually and textually says that Scheme procedure definition page numbers are in italics and page numbers followed by <code>n</code> indicate footnotes. Current physical page 163 says <b>Page numbers for Scheme procedure definitions are included with the generated entries</b> . Source is <code>typ/index.typ:384-386</code> . Future fix should decide whether <code>in-dexter</code> can mark definition locators specially; if not, the note should not promise published behavior, and this limitation should remain explicit.                                                                                                                             |
| C8-006 | Index symbol entries          | <b>BUG</b>                | The published index opens with symbol and notation entries before A: composition circle, flat, derivative notation, partial derivative, <code>df</code> , exterior derivative, tensor selector <code>I_j</code> , Lie derivative, lambda notation, tuple punctuation, quote, <code>#</code> , brackets, parentheses, and Scheme tuple forms. Current physical page 163 starts with section 4, then A/B/C, and lacks the same initial symbol/notation block in the sampled opening page. Published evidence: page 240 visual/text extraction. Current evidence: page 163 visual/text extrac-                                                                                                                                                                 |

|        |                            |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------|----------------------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        |                            |                | tion. Likely repair area is <code>typ/index.typ:13-259</code> for adding explicit symbol/notation entries or a separate manual index prelude if generated matching is insufficient.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| C8-007 | Index code entries         | PARTIAL-EX-ACT | Representative Scheme entries are present in both indexes: published physical page 240 has <code>basis-&gt;1form-basis</code> , <code>basis-&gt;vector-basis</code> , <code>4tuple-&gt;ct</code> , and <code>4tuple-&gt;space</code> ; current physical page 163 has the same visible code entries. However, current entries are generated with raw/code display via <code>fdg-code-index</code> at <code>typ/index.typ:4-8</code> and trigger rules at <code>:261-367</code> , while the published index distinguishes procedure definitions by italic page locators. Current visual page 163 shows highlighted code terms and ordinary generated locators; this is a styling/semantics gap until definition-locator support is implemented or explicitly accepted. |
| C8-008 | Index entry granularity    | BUG            | The generated index is visibly more mechanical and broader than the published index in places: current page 163 shows very dense locator runs for common code terms such as <code>define</code> , <code>differential</code> , <code>down</code> , and <code>compose</code> , while the published page 240 has curated subentries and fewer opening-page locators. This is not just page-number drift; it reflects different index-entry selection and grouping. Future exhaustive pass should build a comparison table with columns <code>term</code> , <code>published entry/subentry</code> , <code>current entry/subentry</code> , <code>published locator class</code> , <code>current locator class</code> , and <code>decision</code> .                        |
| C8-009 | Errata policy              | PROJECT-ADDED  | Errata is intentionally included in the current book after the generated index and is not part of the published PDF target. It should be audited against <code>fdg-book/scheme/org/errata.org</code> and <code>typ/content/errata.typ</code> , not against <code>fdg-book/fdg_book.pdf</code> . Assembly source: converter excludes errata from indexed body at <code>scripts/convert-org-to-typst.mjs:1261-1263</code> , defines <code>errataInclude</code> at <code>:1289</code> , and appends it after the index at <code>:1308</code> . Current physical pages 166-170 render <i>Errata for FDG</i> .                                                                                                                                                            |
| C8-010 | Errata structure           | PARTIAL-EX-ACT | Org and generated Typst agree on the chapter-heading structure in the sampled full file: Org headings at <code>fdg-book/scheme/org/errata.org:1, :3, :9, :20, :28, :70, :79, :131, :161, :218, and :297</code> ; generated headings at <code>typ/content/errata.typ:5, :6, :9, :16, :23, :58, :65, :108, :135, :176, and :248</code> . Current physical page 166 visually shows the title and chapters 1, 3, 4, 5, and 6; pages 167-170 continue through chapter 11. Full sentence-by-sentence errata comparison remains open.                                                                                                                                                                                                                                       |
| C8-011 | Errata code/raw conversion | PARTIAL-EX-ACT | Representative errata code and raw terms convert correctly but with current project styling: Org inline code markers such as <code>=Cartan=</code> , <code>=geodesic-equation-residuals=</code> , and <code>=R2-rect-point=</code> become <code>#raw(lang:"scheme", ...)</code> in <code>typ/content/errata.typ:7-18</code> ; Org source blocks such as <code>fdg-book/scheme/org/errata.org:24-26, :32-40, and :87-103</code> become fenced Scheme blocks at <code>typ/content/errata.typ:19-21, :26-34, and :68-84</code> . Current visual page 166 shows inline raw terms with code styling and syntax-                                                                                                                                                           |

|        |                           |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------|---------------------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        |                           |                | highlighted blocks. This is acceptable project styling unless later errata policy requires plainer prose.                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| C8-012 | Errata content samples    | PARTIAL-EX-ACT | Spot checks preserve high-value errata content: chapter 1 Cartan evaluation-order note from Org <code>fdg-book/scheme/org/errata.org:5-7</code> appears in generated Typst <code>typ/content/errata.typ:7</code> and current physical page 166; chapter 8 torsion example from Org <code>:147-159</code> appears in generated Typst <code>:121-133</code> and current physical page 167; chapter 11 <code>home/the-ether</code> note from Org <code>:305-315</code> appears in generated Typst <code>:251-259</code> and current physical page 170. These are representative only. |
| C8-013 | End-matter ordering       | PROJECT-ADDED  | Current end matter is references, generated index, then errata. Published end matter is references, then index, with no errata. This is intentional, but future release builds should keep the distinction clear: <code>fdg-book.pdf</code> includes project errata, while any strict published-fidelity artifact should be able to omit errata and possibly omit or replace the generated index.                                                                                                                                                                                  |
| C8-014 | Chunk 8 completion marker | UNCHECKED      | Still unchecked for index/errata: every published index term and subterm, alphabetical/symbol sort order, every code/procedure locator, footnote n locators, procedure-definition italic locators, every errata sentence, and every errata code block. Highest-priority follow-up items are symbol-entry coverage, index note/procedure-definition locator semantics, and a machine-readable index comparison table.                                                                                                                                                               |

### Errata Technical Review

This is a mathematical/programmer-confidence pass over `fdg-book/scheme/org/errata.org`, not a proof by executing every Scheme fragment. Scores are out of 5: 5 means likely correct and worth keeping, 3 means plausible but needs a caveat or executable check, and 1 means likely wrong or should be rewritten.

| ID    | Errata source                                                                                         | Score | Assessment and required change                                                                                                                                                                                                                                                                                                          |
|-------|-------------------------------------------------------------------------------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E-001 | <code>errata.org:5-7</code> ; ch.1 page 9 Cartan order                                                | 3     | Historically plausible, but current Org now defines the named <code>Cartan</code> block before <code>geodesic-equation-residuals</code> at <code>chapter001.org:432-449</code> and repeats it with <code>noweb</code> at <code>:459</code> . Rewrite as historical/published-order warning or remove if evaluating current source only. |
| E-002 | <code>errata.org:11-13</code> ; ch.3 v assumes <code>R2-&gt;R</code>                                  | 4     | Likely correct as chapter-isolation setup. The vector-field example uses <code>R2-&gt;R</code> at <code>chapter003.org:146-150</code> ; that signature is introduced earlier, not locally. Keep, but phrase as <code>when evaluating chapter 3 independently</code> .                                                                   |
| E-003 | <code>errata.org:15-16</code> ; ch.3 listing assumes <code>R2-rect-point</code>                       | 4     | Likely correct as setup hygiene. Chapter 3 examples use <code>R2-rect-point</code> in early listings, and the current regression suite preserves a repair around defining it. Keep, but identify the exact listing in a future executable pass.                                                                                         |
| E-004 | <code>errata.org:18</code> ; ch.3 <code>omega</code> already defined                                  | 2     | Probably true but not a real error by itself: redefining <code>omega</code> in an interactive Scheme session is usually harmless unless the old binding is needed. Rewrite as a low-severity note or drop from errata proper.                                                                                                           |
| E-005 | <code>errata.org:22-26</code> ; ch.4 install <code>R2-rect</code>                                     | 3     | Plausible if the reader starts chapter 4 cold. It is less compelling in a continuous session because <code>R2-rect</code> is used heavily earlier. Keep as setup requirement, not mathematical correction.                                                                                                                              |
| E-006 | <code>errata.org:30-41</code> ; ch.5 needs <code>R3-rect-point</code>                                 | 4     | Likely correct. Chapter 5 uses <code>R3-rect-point</code> in examples after defining coordinates, and the suggested point definition is standard. Keep.                                                                                                                                                                                 |
| E-007 | <code>errata.org:43-58</code> ; ch.5 cylindrical <code>z</code> overwrites rectangular <code>z</code> | 2     | This is a stylistic/environment concern, not clearly a bug: same named coordinate functions for the same geometric coordinate are often inten-                                                                                                                                                                                          |

|       |                                                              |   |                                                                                                                                                                                                                                                                        |
|-------|--------------------------------------------------------------|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |                                                              |   | tional in this code style. Keep only as optional caution, or move out of errata.                                                                                                                                                                                       |
| E-008 | errata.org:60-61; ch.5 alpha/beta assume R2->R               | 4 | Likely correct as setup hygiene. Chapter 5 defines alpha and beta using R2->R at chapter005.org:359-360; keep with chapter-isolation wording.                                                                                                                          |
| E-009 | errata.org:63-68; ch.5 needs R2-rect-point                   | 5 | Likely correct and actionable. Chapter 5 uses R2-rect-point later, and the suggested definition is appropriate. Keep.                                                                                                                                                  |
| E-010 | errata.org:72-77; ch.6 S^2 vs S^2-type                       | 5 | Likely correct hard runtime bug. chapter006.org:169 has (make-manifold S^2 2 3), while other working examples use S^2-type. Keep.                                                                                                                                      |
| E-011 | errata.org:81-106; ch.7 page 92 result mismatch              | 3 | Plausible, but this needs an executable scmutils transcript before being treated as settled. The diagnosis is detailed enough to keep, but mark as needs reproduction and include scmutils version/date.                                                               |
| E-012 | errata.org:108-119; ch.7 J should be circular                | 5 | Likely correct. The local vector field is named circular at chapter007.org:971, but the later polar-Cartan call uses J at :1001. Keep; also note nearby source has d/x, which may be another typo for d/dx.                                                            |
| E-013 | errata.org:121-126; ch.7 S2-Christoffel needs coordinates    | 4 | Mostly correct as setup hygiene. The chapter defines S2-spherical at chapter007.org:1138-1139; a reader evaluating only the Christoffel block would need the coordinate setup. Keep with context.                                                                      |
| E-014 | errata.org:128-129; ch.7 sphere uses S^2                     | 5 | Likely correct hard runtime bug. chapter007.org:1137 has (make-manifold S^2 2 3), matching the chapter 6 issue. Keep.                                                                                                                                                  |
| E-015 | errata.org:133-138; ch.8 needs S2-spherical                  | 4 | Likely correct as setup hygiene. Chapter 8 begins with code using sphere-Cartan, dphi, d/dtheta, and S2-spherical context from earlier chapters. Keep but classify as setup dependency.                                                                                |
| E-016 | errata.org:140-145; ch.8 define omega                        | 3 | Conceptually right that the later prose needs an arbitrary one-form, but the suggested procedure name is wrong for this codebase: use literal-1form-field, not literal-oneform-field. Also quote the symbol: (define omega (literal-1form-field 'omega S2-spherical)). |
| E-017 | errata.org:147-159; ch.8 torsion example needs f             | 5 | Likely correct. The torsion-vector example at chapter008.org:467-488 uses an inline literal function, while the later errata-targeted example uses free f. Keep; suggested definition should quote the symbol: (define f (literal-manifold-function 'f S2-spherical)). |
| E-018 | errata.org:163-166; ch.9 raise lacks procedure->vector-field | 4 | Likely correct as API consistency. lower wraps with procedure->1form-field; raise at chapter009.org:68-73 returns a lambda directly. It may still work by operator conventions, so keep as worth checking/modernizing, not definitely fatal.                           |
| E-019 | errata.org:168-174; ch.9 page 136 setup                      | 4 | Likely correct as setup hygiene: page 136 code depends on S2-spherical and S2-basis. Keep.                                                                                                                                                                             |
| E-020 | errata.org:176-180; ch.9 simplifier 3/2 denominator          | 3 | Plausible but version-sensitive. This is a computational-reproducibility note, not a book math error until tested against a pinned scmutils revision. Keep with exact scmutils version and transcript requirement.                                                     |
| E-021 | errata.org:183-190; ch.9 spacetime setup                     | 5 | Likely correct. Section 9.3 uses spacetime-rect and spacetime-rect-basis without local setup. Keep.                                                                                                                                                                    |
| E-022 | errata.org:192-198; ch.9 V undefined                         | 5 | Likely correct. The Newton metric examples pass V into Newton-metric; an explicit literal function definition is appropriate. Keep.                                                                                                                                    |
| E-023 | errata.org:200-216; ch.9 Newton-metric result                | 3 | Mathematically plausible, but should be re-run before being asserted as a correction. It may be a weak-field/leading-term presentation rather than an error. Keep as needs explanation or leading-term note.                                                           |
| E-024 | errata.org:220-232; ch.10 SR-basis                           | 5 | Likely correct hard setup bug. Chapter 10 defines SR-vector-basis at chapter010.org:394 but calls SR-basis at :414 and later. Keep; the suggested SR-vector-basis simplification is reasonable.                                                                        |

|       |                                                           |   |                                                                                                                                                                                                                                                                                                                                                     |
|-------|-----------------------------------------------------------|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E-025 | errata.org:234-292;<br>ch.10 c <sup>2</sup> normalization | 3 | Conceptually important but too broad to treat as a simple erratum. The prose metric includes c <sup>2</sup> at <a href="#">chapter010.org:351-355</a> , while the program metric at <a href="#">:362-366</a> is normalized in ct coordinates. Keep as a design/notation clarification, but test every proposed substitution before publishing them. |
| E-026 | errata.org:294-295;<br>ch.10 eta-inverse un-<br>defined   | 5 | Likely correct hard runtime bug. Force calls eta-inverse at <a href="#">chapter010.org:725-733</a> ; no local definition was found. Keep and add a concrete definition or replace with the inverse metric accessor used elsewhere.                                                                                                                  |
| E-027 | errata.org:299-302;<br>ch.11 slow simplifica-<br>tion     | 2 | Probably true for some environments but not really a book erratum unless the book promises practical evaluation. Keep as performance/reproducibility note only with machine, scmutils version, and timeout details.                                                                                                                                 |
| E-028 | errata.org:305-315;<br>ch.11 home frame                   | 3 | Plausible but under-evidenced in this repo: the chapter uses home at <a href="#">chapter011.org:641-652</a> , but the claimed library binding the-ether was not found in the checked Org/scm files. Keep only if verified against the installed scmutils library; otherwise suggest defining home directly with frame-maker.                        |

Chunk 8 evidence commands:

```
nl -ba typ/index.typ | sed -n '1,430p'
nl -ba typ/content/errata.typ | sed -n '1,340p'
nl -ba fdg-book/scheme/org/errata.org | sed -n '1,340p'
nl -ba scripts/convert-org-to-typst.mjs | sed -n '1248,1312p'
pdftotext -f 163 -l 170 fdg-book.pdf -
pdftotext -f 240 -l 249 fdg-book/fdg_book.pdf -
pdftoppm -f 163 -singlefile -png -r 130 fdg-book.pdf /tmp/fdg-current-index-p163
pdftoppm -f 166 -singlefile -png -r 130 fdg-book.pdf /tmp/fdg-current-errata-p166
pdftoppm -f 240 -singlefile -png -r 130 fdg-book/fdg_book.pdf /tmp/fdg-published-index-p240
```

## Exhaustive Audit Plan

For each region, future runs should append entries in this form:

```
ID:
Region:
Source span:
Published PDF span:
Typst span:
Status: EXACT | PARTIAL-EXACT | ACCEPTED-STYLISTIC | SOURCE-DRIFT | PDF-DERIVED-REPAIR | PROJECT-ADDED
| BUG | UNCHECKED
Evidence:
Decision:
Follow-up:
Commit(s):
```

The exhaustive audit should proceed in bounded chunks. Each chunk should update this file with concrete entries and leave any unfinished work explicitly marked **PARTIAL** or **UNCHECKED**.

## Evidence Preservation Plan

Many current audit rows cite `fdg-book.pdf` physical pages as they existed during the audit. Fixing earlier chunks will legitimately reflow later pages, so those current-PDF page anchors are not stable evidence unless we preserve the build that produced them.

| Rule                  | Plan                                                                                                                                                                                    |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Snapshot before fixes | Before beginning a fix pass for a chunk, copy the current rendered artifacts into a dated audit snapshot directory, for example <code>audit-snapshots/2026-07-03-before-fixes/</code> , |

|                       |                                                                                                                                                                                                                                                                                                                                      |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | including <code>fdg-book.pdf</code> , <code>typ/audit.pdf</code> , and any rasterized pages used for visual evidence. Record the commit hash and build commands in a short manifest.                                                                                                                                                 |
| Stable citations      | Existing rows that mention <b>current physical page N</b> should be read as <b>current PDF at the audit snapshot for that row</b> , not necessarily the latest rebuilt <code>fdg-book.pdf</code> . Future rows should name the snapshot when citing unstable current pages, for example <b>current snapshot 2026-07-03 page 82</b> . |
| Published PDF anchors | Published physical page anchors remain stable because <code>fdg-book/fdg_book.pdf</code> is an upstream artifact and should not be regenerated during Typst fixes. Keep using published pages as canonical visual targets.                                                                                                           |
| Prefer source anchors | Every actionable row should include Org, generated Typst, and converter/helper line anchors wherever possible. Source anchors survive reflow better than current-PDF page anchors and should be the primary repair coordinates.                                                                                                      |
| After each fix        | After applying fixes, rebuild <code>fdg-book.pdf</code> and add a new verification note or row that records the new current page(s). Do not rewrite old evidence in a way that obscures what the previous row observed; instead mark it superseded or add <b>verified after fix</b> evidence.                                        |
| Automation target     | A later script should create snapshots, record <code>git rev-parse HEAD</code> , run <code>pdfinfo</code> , optionally rasterize cited pages, and produce a manifest that can be referenced from this audit. Until that exists, do the snapshot step manually before each fix chunk.                                                 |

| Chunk | Scope                           | Deliverable                                                                                          | Notes                                                                                                                      |
|-------|---------------------------------|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| 1     | Front matter and global rules   | Cover/title/copyright/quote, contents, page numbering, draft/book differences, errata/index policy.  | Mostly accepted layout/style differences. Confirm no content is dropped or obscured.                                       |
| 2     | References and citations        | All 21 bibliography entries plus all citation callouts.                                              | High priority because references affect the whole text. Numbering/order now matches samples; formatting still needs audit. |
| 3     | Preface, prologue, chapters 1-2 | First full prose/math/code/footer workflow.                                                          | These regions already had deeper prior passes, so they are ideal for validating the audit method.                          |
| 4     | Chapters 3-5                    | Section-by-section audit, including equations, code/results, figures through 5.1.                    | Covers vector fields, basis fields, and integration.                                                                       |
| 5     | Chapters 6-8                    | Equation-number audit, code/result alignment, figures 6.1 and 7.1, known chapter 7 fix verification. | Math/code density is high; keep entries section-scoped before drilling into individual blocks.                             |
| 6     | Chapters 9-11                   | Metrics, Hodge star/electrodynamics, special relativity, chapter 11 PDF-derived repairs.             | Explicitly verify equations 11.1-11.14 and 11.20-11.21 against the published PDF.                                          |
| 7     | Appendices A-C                  | Code/prose fidelity, citations, Appendix C tensor equation/reference checks.                         | Appendices are source-heavy and benefit from a separate pass.                                                              |
| 8     | Index and errata                | Generated index comparison policy; errata compared against Org/Typst source.                         | Index page locators cannot match exactly because page geometry intentionally differs.                                      |

Recommended workflow for each chunk:

1. Extract normalized text from the published PDF, Org source, generated Typst source, and current `fdg-book.pdf`.

2. Compare first at section level, then split mismatching sections into prose, equation, code, figure, footnote, citation, and index/reference entries.
3. Classify each mismatch as EXACT, ACCEPTED-STYLISTIC, SOURCE-DRIFT, PDF-DERIVED-REPAIR, PROJECT-ADDED, BUG, or UNCHECKED.
4. For every BUG, SOURCE-DRIFT, or PDF-DERIVED-REPAIR row, include enough repair coordinates for a future maintainer to start efficiently: Org file/line, generated Typst file/line, published/current physical PDF page, visual-vs-text-extraction status, and likely helper/converter file to inspect.
5. Before fixing a chunk, snapshot the current rendered PDFs and record the snapshot name in the audit entry or chunk evidence notes, because subsequent fixes can reflow `fdg-book.pdf` and invalidate bare current-page references.
6. Fix clear bugs immediately when they are local and low risk; otherwise leave an audit entry with evidence and follow-up.
7. Rebuild `fdg-book.pdf` and `typ/audit.pdf`, then record verification evidence in this file without deleting the pre-fix evidence trail.

## Settled Decisions and Remaining Questions

- Settled: letter layout is intentional.
- Settled: errata is intentional project end matter.
- Settled: `typ/main.pdf` and `fdg-book.pdf` should differ only by cover/index material.
- Settled: this audit lives as a standalone Typst document.
- Settled: published footnote numbering is a fidelity requirement.
- Settled: generated index entries should be compared by vocabulary, grouping, ordering, symbol/code styling, footnote locator notation, and procedure-definition locator styling; exact page-number equality is not required while page geometry differs intentionally.
- Settled: current-PDF page anchors are unstable across fixes and must be tied to audit snapshots or superseded by post-fix verification rows.
- Direction: redraw figures natively in CeTZ after or alongside the figure audit, replacing imported PDF crops where practical.
- Remaining: whether current generated IEEE-style bibliography formatting is acceptable now that numbering/order matches.

## Commit Trail

| Commit  | Date       | Message                                                           |
|---------|------------|-------------------------------------------------------------------|
| 693bae8 | 2026-07-03 | cleanup                                                           |
| 9512cbe | 2026-07-03 | draft as default + makefile                                       |
| 413cbff | 2026-07-02 | fixes                                                             |
| a1006ad | 2026-07-02 | fmt                                                               |
| c9e554f | 2026-07-02 | add license, typos                                                |
| 650114f | 2026-07-02 | header                                                            |
| c7b3523 | 2026-07-02 | add cover                                                         |
| 289a9dc | 2026-07-02 | updates                                                           |
| 20c5627 | 2026-07-02 | fmt links                                                         |
| e12283d | 2026-07-02 | fmt                                                               |
| 4eef47d | 2026-07-02 | fixes                                                             |
| eb71a72 | 2026-07-02 | refs                                                              |
| 8ed9337 | 2026-07-02 | add references                                                    |
| 4460b9f | 2026-07-02 | fix                                                               |
| 3128ce4 | 2026-07-02 | cleanup                                                           |
| 67eefdb | 2026-07-02 | init                                                              |
| cfa1f68 | 2026-07-02 | Merge commit e1481748be83c493edfe54271d2068577aef1e2d as fdg-book |
| e148174 | 2026-07-02 | Squashed fdg-book/ content from commit 9beb699                    |
| 9edbac7 | 2026-07-02 | Initial commit                                                    |